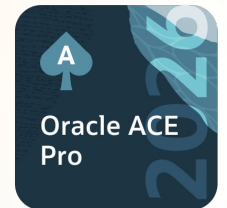


# Oracle SQLcl: The Free Tool Every PL/SQL and APEX Developer Needs

Tuesday, Jun 09, 2026 at 11:15 AM - 12:15 PM EDT  
Ascend 2026, Orlando, June 8-11.2026



**Rafał Grzegorzczuk**  
Senior APEX Developer





# Oracle SQLcl (SQL Developer Command Line)

- SQLcl is SQL Developer just with CLI vs a GUI
- Over 100 of useful commands
- First, early adopter version was mentioned around 2014/2015
- "SQL\*PLus++" , "SDSQL" \*
- Modern replacement of SQL Plus
- Latest version 26.1.2 (May 2026)

<https://www.oracle.com/pl/database/sqldeveloper/technologies/sqlcl/>  
<https://krisrice.io/2014-12-11-what-is-sdsq/>

```
ACCEPT
ALIAS
APEX
APPEND
AQ
ARBORI
ARCHIVE_LOG
ARGUMENT
AWR
BACKGROUND
BLOCKCHAIN_TABLE
BREAK
BRIDGE
BTITLE
CD
CERTIFICATE
CHANGE
CLEAR
CLOUDSTORAGE
CODESCAN
COLUMN
COMPUTE
CONNECT
CONNMGR
COPY
CS
CTAS
DATAPUMP
:
```

## About me

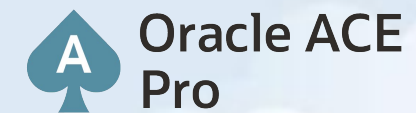
- Kielce, Poland - 200km (125 miles) south from Warsaw
- Oracle APEX and PL/SQL Developer
- Worldwide speaker since 2021
- Oracle ACE Pro

Find me at:

[rgrzegorczyk@pretius.com](mailto:rgrzegorczyk@pretius.com)

[rgrzegorczyk](#) 

<https://rafal.hashnode.dev/>



**Rafał Grzegorczyk**





# Providing what you **need**, not what you **want**

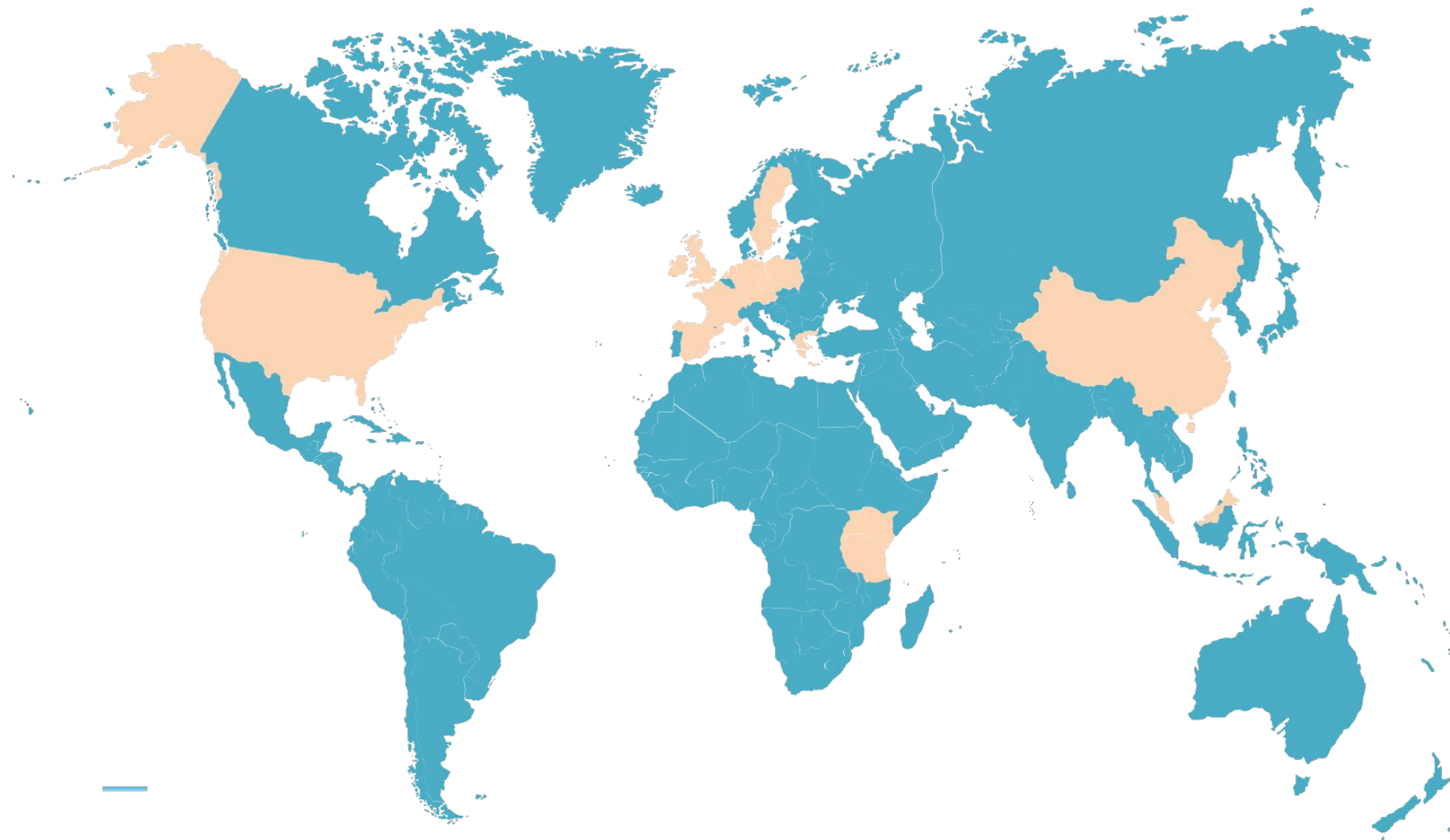


Scan to meet us  
Pretius.com

**300+**  
People

**60+**  
APEX Devs

**7**  
Oracle ACEs



**70+**  
Customers

**4**  
Continents

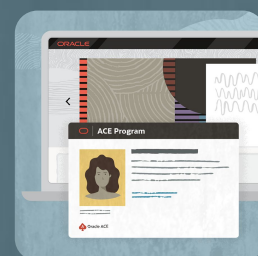
**2**  
Warsaw, PL  
London, UK





# Tech Superstars Unite

Get worldwide recognition as an Oracle ACE



Oracle.com profile page



Swag, certification exam credit & event passes



Exclusive content



Networking events



Your own Oracle cloud account



Travel support



X @oracleace

in [Linkedin.com/groups/72183](https://www.linkedin.com/groups/72183)

🦋 @oracleace.bsky.social

Learn more at:  
[ace.oracle.com](https://ace.oracle.com)





## Audience Q&A





## Which tool do you use?







**GitHub - rgrzegorzcyk/conferences: A place  
for public content related to my  
presentations at various conferences**

github.com

<https://github.com/rgrzegorzcyk/conferences>



## SQLcl features I find most useful:

1. Model Context Protocol (MCP) and effortless Skills installation
2. Import/export data from/to files
3. Data Pump - export/import schemas
4. APEX export (APEXlang included)
5. CODESCAN - PL/SQL quality and performance checks
6. Projects - Database Application CI/CD
7. Liquibase
8. Schema DIFF

# SQLcl Installation



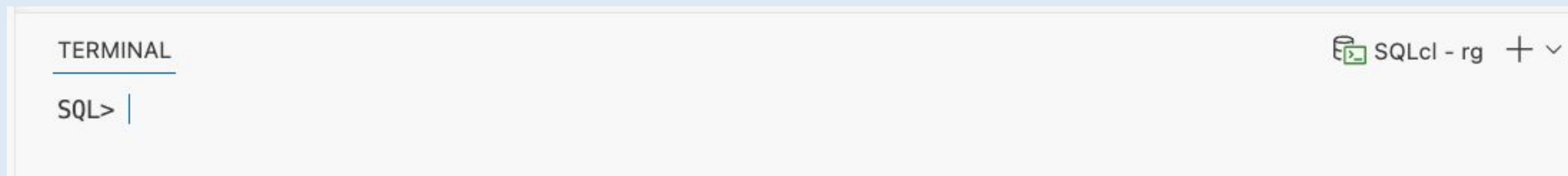
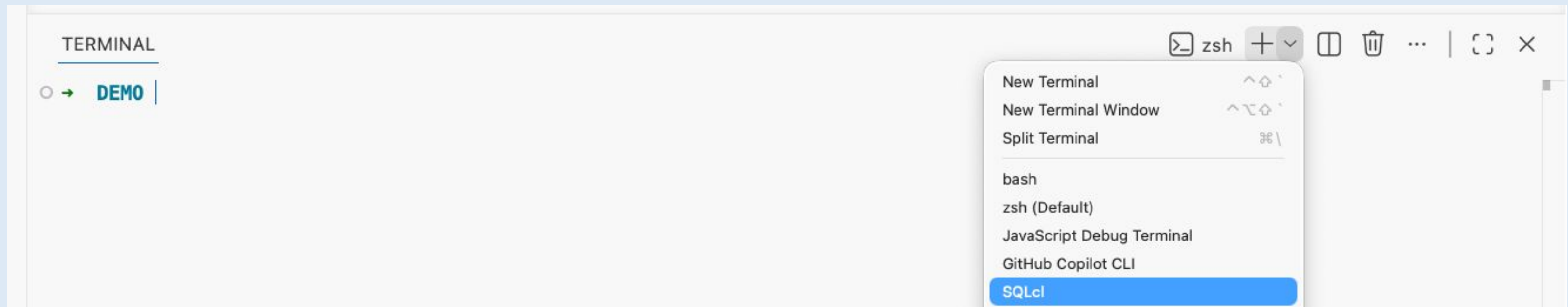
Scan to ask a question





Do you have SQL Developer Extension in Visual Studio Code?

**SQLcl is already there**





For standalone version go  
to **[www.oracle.com/sqlcl](http://www.oracle.com/sqlcl)**

## SQLcl Downloads

Download the latest version of SQLcl now

[Download](#)

[Get Started](#) [Release Notes](#) [Documentation](#) [Forums](#)

## Download

Product Versions

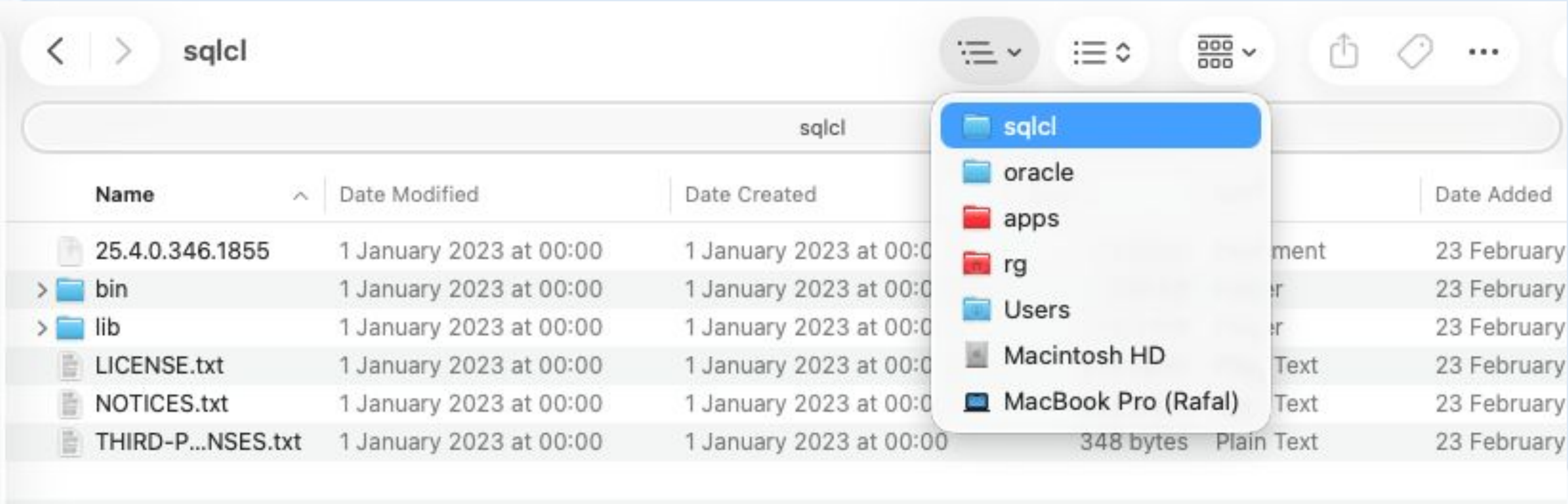
[25.4.2.044.1837](#)

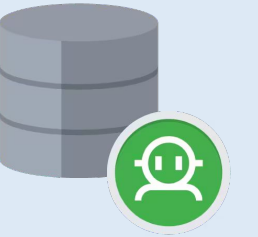
Related Resources

- MD5: fb4536a793df5e03dc52d9078221f798
- SHA1: dd0d80afe1ac8494565924b4e29c1949d1c31792
- [Oracle Free Use License](#)



Unzip downloaded file and add to chosen folder





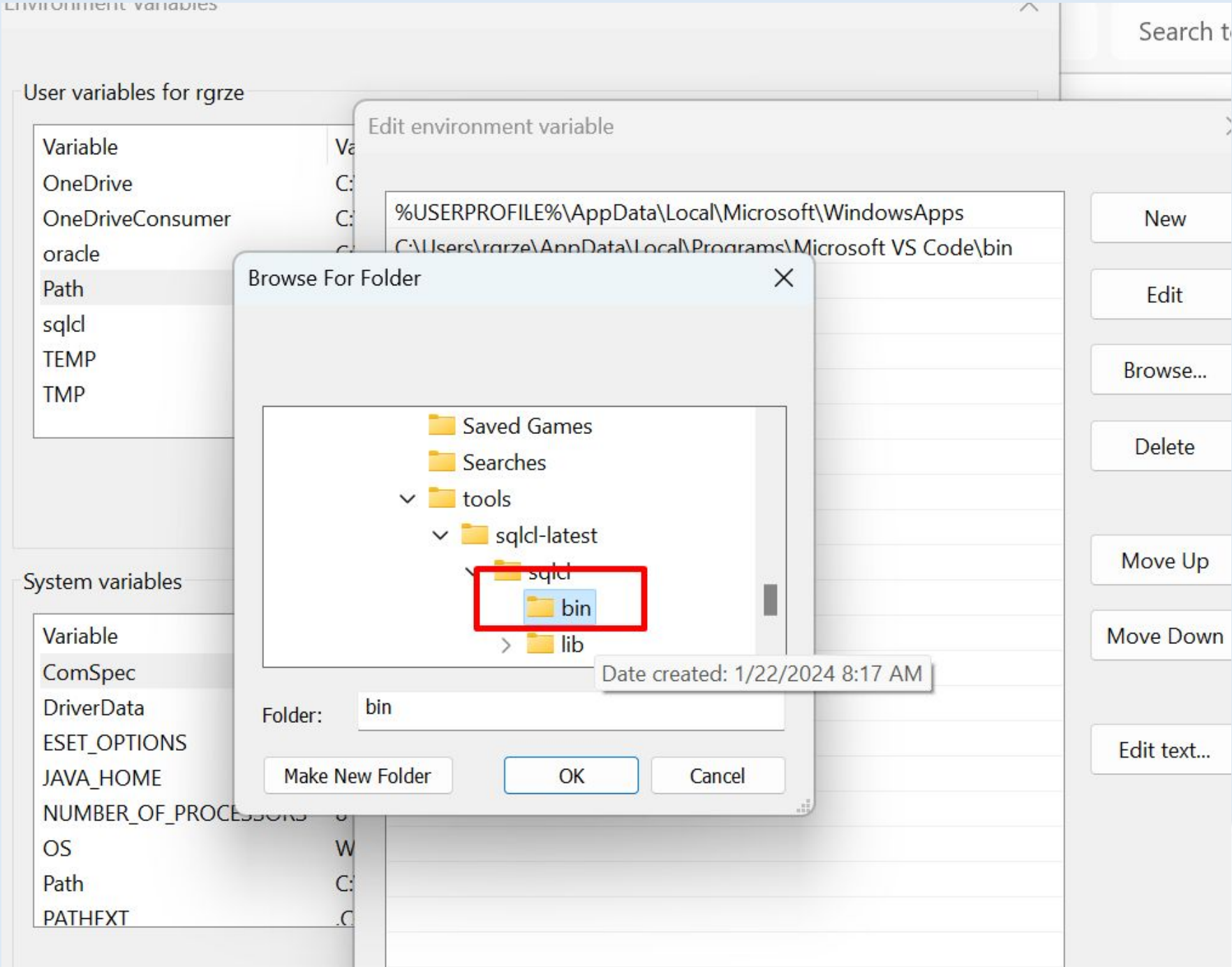
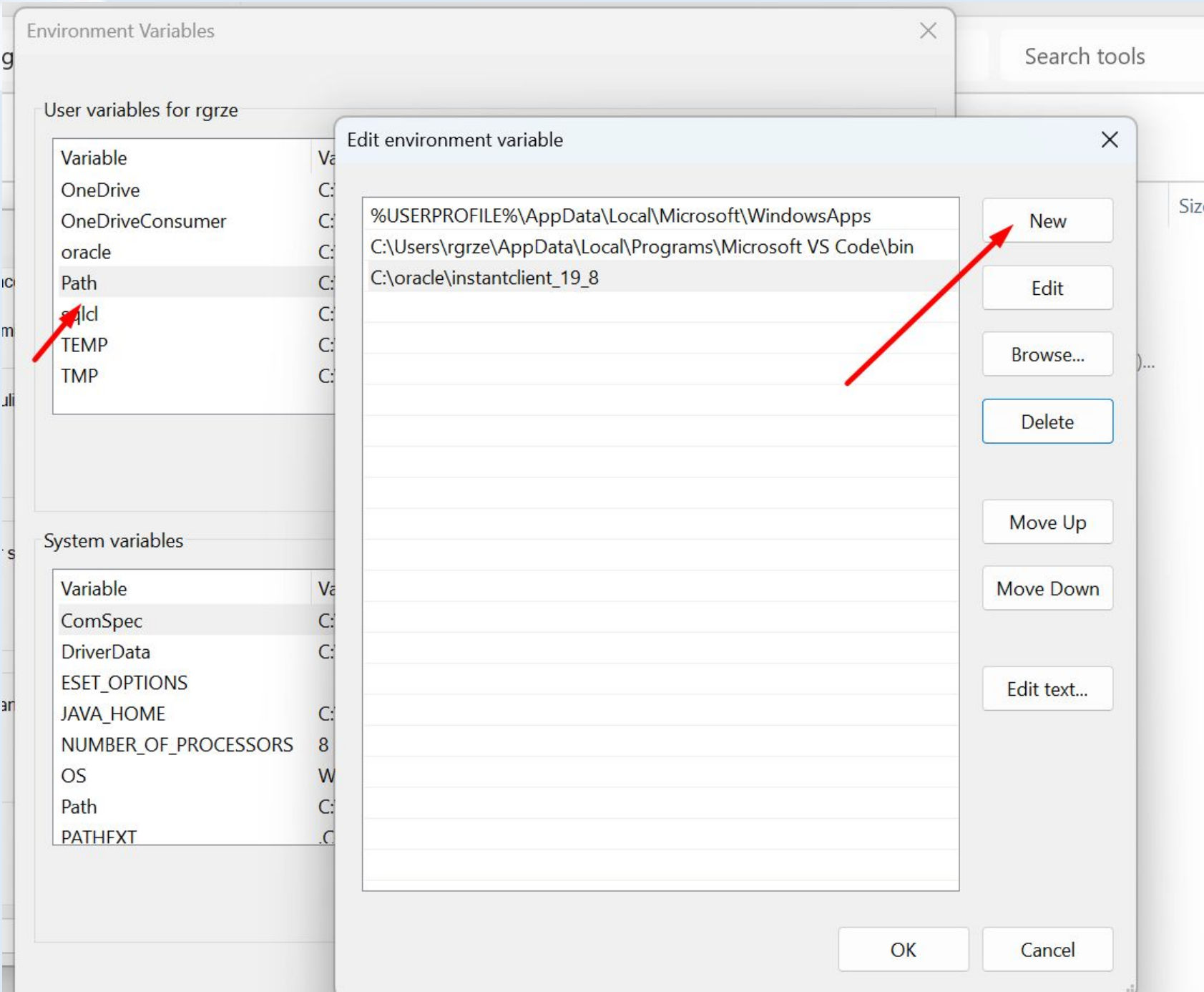
Add SQLcl "bin" folder to system path (**MacOS**)

Users/<your\_user>/.zshrc (or .bash\_profile)

```
export PATH="/Users/rg/apps/oracle/sqlcl/bin:$PATH"
```



Add SQLcl “bin” folder to system path (Windows)





Verify installation (JAVA is required)

> java -version

#### TERMINAL

```
● → DEMO java -version
java version "25.0.2" 2026-01-20 LTS
Java(TM) SE Runtime Environment (build 25.0.2+10-LTS-69)
Java HotSpot(TM) 64-Bit Server VM (build 25.0.2+10-LTS-69, mixed mode, sharing)
○ → DEMO
```

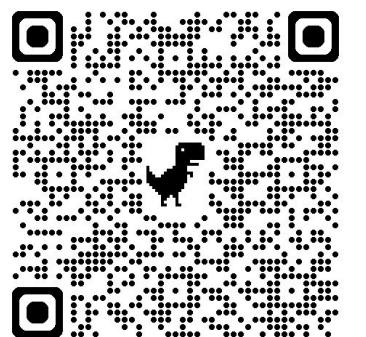
> sql -version

#### TERMINAL

```
● → DEMO sql -version
SQLcl: Release 25.4.0.0 Production Build: 25.4.0.346.1855
○ → DEMO
```

Supported Java Versions -> 17, 21

More about installation [here](#) ->



# SQLcl connect to database



Scan to ask a question





Visual Studio Code SQL Developer Extension **(easiest way)**

EXTENSIONS: INSTALLED

@installed SQL Developer

**Oracle SQL Developer ...** 117ms  
End-to-end development tools fo...  
Oracle Corporation

Extension: Oracle SQL Developer Extension for VSCode

### Oracle SQL Developer Extension for VSCode

Oracle Corporation [oracle.com](#) | 758,601 | (115)

End-to-end development tools for your SQL and PL/SQL applications.

Disable

Uninstall

☒ Auto Update

DETAILS

FEATURES

CHANGELOG

## Oracle SQL Developer Extension for VS Code

### About the Oracle SQL Developer extension for VS Code

The gold standard for Oracle Database development and administration is now seamlessly integrated into Visual Studio Code

#### Installation

|              |                      |
|--------------|----------------------|
| Identifier   | oracle.sql-developer |
| Version      | 25.4.1               |
| Last Updated | 2 weeks ago          |
| Size         | 493.92MB             |





Visual Studio Code SQL Developer Extension **(easiest way)**

SQL DEVELOPER

SQL SNIPPETS

CONNECTIONS

customers

priv

demo

Connection: dev\_demo

Update Connection dev\_demo

Connection Name

dev\_demo

User Info

Proxy User

Authentication Type

Default

Role

Default

Username

DEMO

Password

.....

Save Password

Connection Type

Cloud Wallet

Details

Proxy

Advanced

Configuration File

Choose File

Service

DEV\_MEDIUM

Cancel

Test

Delete

Connect

Save

Details

Proxy

Advanced

Configuration File

Choose File

Service

DEV\_MEDIUM

Cancel

Test

Test passed for connection: dev\_demo

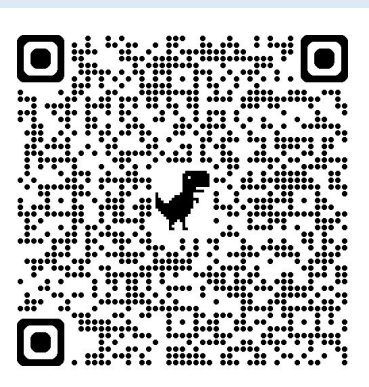
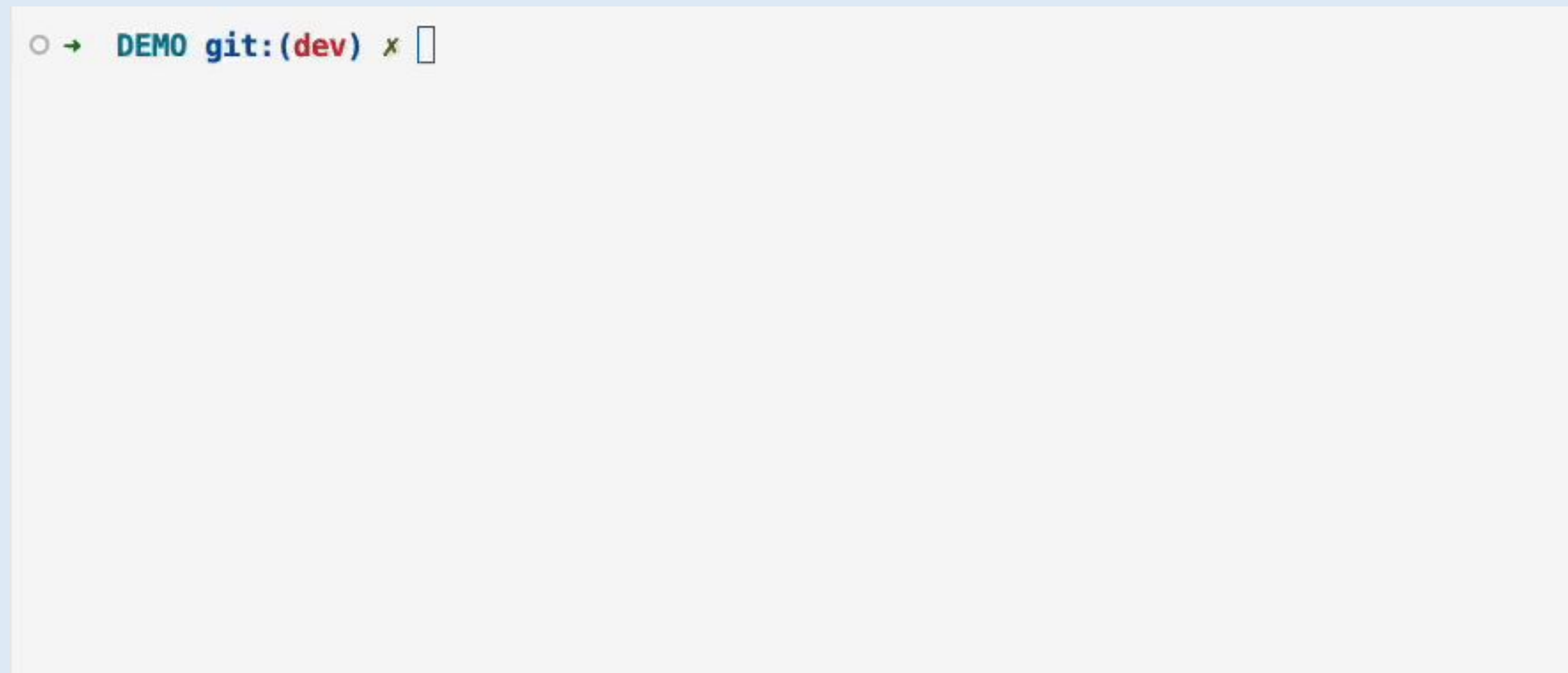
Source: Oracle SQL Developer Extension for VSCode

Cancel



Using SQLcl commands **(also easy)**

> conn -save



<- More about connections [here](#)



```
> sql -name <connection_name>
```

#### TERMINAL

```
○ → DEMO sql -name dev_demo
```

```
SQLcl: Release 25.4 Production on Tue Mar 10 10:41:56 2026
```

```
Copyright (c) 1982, 2026, Oracle. All rights reserved.
```

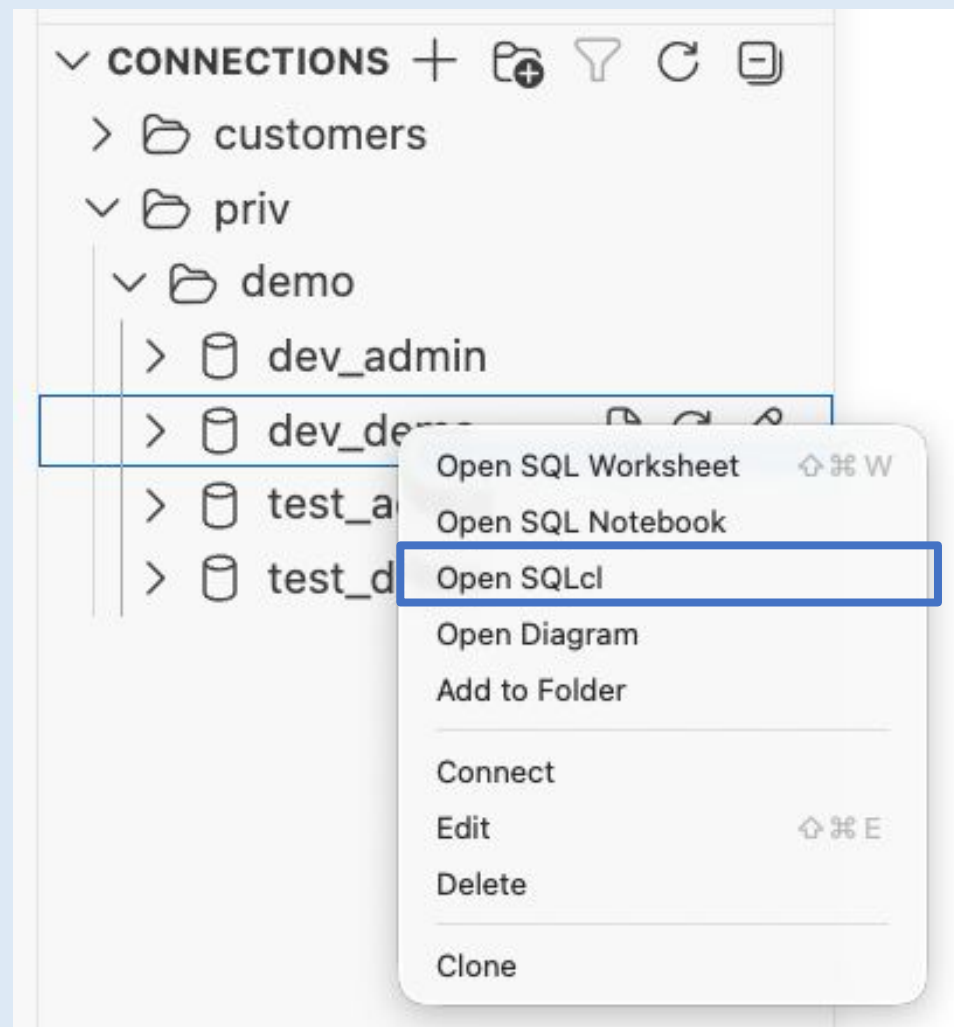
```
Connected to:
```

```
Oracle AI Database 26ai Enterprise Edition Release 23.26.1.2.0 – Production  
Version 23.26.1.2.0
```

```
SQL>
```



## Using VS Code SQL Developer extension

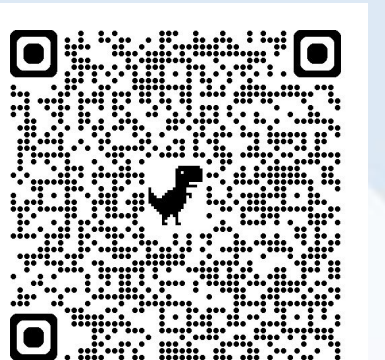


```
TERMINAL
SQLcl - dev_demo - rg + v

Copyright (c) 1982, 2026, Oracle. All rights reserved.
Connected to:
Oracle AI Database 26ai Enterprise Edition Release 23.26.1.2.0 - Production
Version 23.26.1.2.0

SQL>
```

More about connections [here](#) ->

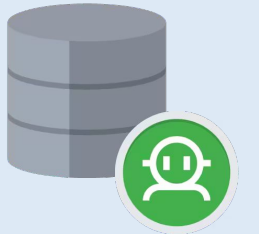




# Export data using SQLcl



Scan to ask a question



> set sqlformat insert

```
SQL> set sqlformat insert
SQL> select name, country, rating from eba_demo_ig_people fetch first 10 rows only;
REM INSERTING into EBA_DEMO_IG_PEOPLE
SET DEFINE OFF;
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Keizo Obuchi','Japan',22.475646);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Antonín Novotný','Czech Republic',22.474758);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Shivaji','India',22.474212);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Frank Rijkaard','Netherlands',22.473873);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Richard Pryor','United States',22.473754);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Nico','Germany',22.473584);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Alexander II of Scotland','United Kingdom',22.473236);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Jacques Anquetil','France',22.472601);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Hideki Shirakawa','Japan',22.47257);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Chet Atkins','United States',22.472027);
```

10 rows selected.





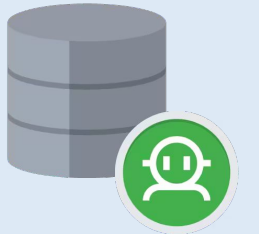
```
> set sqlformat json
```

```
SQL> set sqlformat json
SQL> select name, country, rating from eba_demo_ig_people fetch first 10 rows only;
{"results":[{"columns":[{"name":"NAME","type":"VARCHAR2"}, {"name":"COUNTRY","type":"VARCHAR2"}, {"name":"RATING","type":"NUMBER"}], "items":
[
{"name":"Giuseppe Farina","country":"Italy","rating":22.153414}
,{"name":"Geena Davis","country":"United States","rating":22.152783}
,{"name":"Bengt I. Samuelsson","country":"Sweden","rating":22.152342}
,{"name":"Susan Hayward","country":"United States","rating":22.151325}
,{"name":"Grand Duchess Maria Nikolaevna of Russia","country":"Russia","rating":22.151002}
,{"name":"Chang Myon","country":"South Korea","rating":22.149127}
,{"name":"Vladimir Voronin","country":"Moldova","rating":22.148992}
,{"name":"P. W. Botha","country":"South Africa","rating":22.148361}
,{"name":"Hans Sloane","country":"United Kingdom","rating":22.146437}
,{"name":"Howard Aiken","country":"United States","rating":22.14543}
]
]
}}
10 rows selected.
```

```
> select /*insert*/ (...)
```

```
> select /*csv*/ (...)
```

```
> select /*json*/ (...) also works
```



```
> set sqlformat csv
> spool [file_name]
> spool off
```

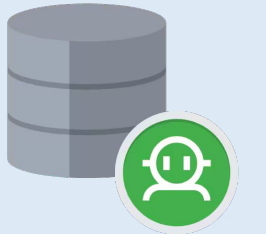
```
SQL> set feedback off
SQL> set sqlformat csv
SQL> spool my_csv_file.csv
SQL> select name,country,rating from eba_demo_ig_people fetch first 10 rows only;
"NAME","COUNTRY","RATING"
"Keizo Obuchi","Japan",22.475646
"Antonín Novotný","Czech Republic",22.474758
"Shivaji","India",22.474212
"Frank Rijkaard","Netherlands",22.473873
"Richard Pryor","United States",22.473754
"Nico","Germany",22.473584
"Alexander II of Scotland","United Kingdom",22.473236
"Jacques Anquetil","France",22.472601
"Hideki Shirakawa","Japan",22.47257
"Chet Atkins","United States",22.472027
SQL> spool off;
```

my\_csv\_file.csv X

my\_csv\_file.csv > data

|    |                                             |
|----|---------------------------------------------|
| 1  | "NAME","COUNTRY","RATING"                   |
| 2  | "Johann Sebastian Bach","Germany",30.167197 |
| 3  | "Ludwig van Beethoven","Germany",30.109331  |
| 4  | "Cleopatra VII of Egypt","Egypt",30.036677  |
| 5  | "Galileo Galilei","Italy",29.96003          |
| 6  | "Euclid","Unknown",29.884558                |
| 7  | "Nero","Italy",29.859917                    |
| 8  | "Karl Marx","Germany",29.842897             |
| 9  | "Marco Polo","Italy",29.771886              |
| 10 | "Cicero","Italy",29.742594                  |
| 11 | "Vincent van Gogh","Netherlands",29.738438  |
| 12 |                                             |





```
> set sqlformat json
> spool [file_name]
> spool off
```

```
SQL> set sqlformat json
SQL> spool my_json.json
SQL> select name, country, rating from eba_demo_ig_people fetch first 10 rows only;
```

```
{
  "results": [
    {
      "columns": [
        { "name": "NAME", "type": "VARCHAR2" },
        { "name": "COUNTRY", "type": "VARCHAR2" },
        { "name": "RATING", "type": "NUMBER" }
      ],
      "items": [
        { "name": "Johann Sebastian Bach", "country": "Germany", "rating": 30.167197 },
        { "name": "Ludwig van Beethoven", "country": "Germany", "rating": 30.109331 },
        { "name": "Cleopatra VII of Egypt", "country": "Egypt", "rating": 30.036677 },
        { "name": "Galileo Galilei", "country": "Italy", "rating": 29.96003 },
        { "name": "Euclid", "country": "Unknown", "rating": 29.884558 },
        { "name": "Nero", "country": "Italy", "rating": 29.859917 },
        { "name": "Karl Marx", "country": "Germany", "rating": 29.842897 },
        { "name": "Marco Polo", "country": "Italy", "rating": 29.771886 },
        { "name": "Cicero", "country": "Italy", "rating": 29.742594 },
        { "name": "Vincent van Gogh", "country": "Netherlands", "rating": 29.738438 }
      ]
    }
  ]
}
```

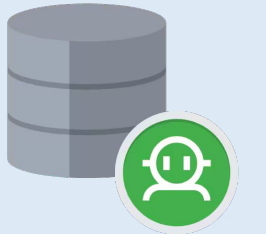
```
SQL> spool off;
```

{} my\_json.json X

{} my\_json.json > ...

```
1  { "results": [ { "columns": [ { "name": "NAME", "type": "VARCHAR2" }, { "name": "CO
2  [
3    { "name": "Giuseppe Farina", "country": "Italy", "rating": 22.153414 }
4    , { "name": "Geena Davis", "country": "United States", "rating": 22.152783 }
5    , { "name": "Bengt I. Samuelsson", "country": "Sweden", "rating": 22.152342 }
6    , { "name": "Susan Hayward", "country": "United States", "rating": 22.151325 }
7    , { "name": "Grand Duchess Maria Nikolaevna of Russia", "country": "Russia
8    , { "name": "Chang Myon", "country": "South Korea", "rating": 22.149127 }
9    , { "name": "Vladimir Voronin", "country": "Moldova", "rating": 22.148992 }
10   , { "name": "P. W. Botha", "country": "South Africa", "rating": 22.148361 }
11   , { "name": "Hans Sloane", "country": "United Kingdom", "rating": 22.146437 }
12   , { "name": "Howard Aiken", "country": "United States", "rating": 22.14543 }
13   ] ] ]
14
```





- > set loadformat csv
- > unload sample\_restaurant

```
SQL> set loadformat csv
SQL> unload sample_restaurant
```

```
format csv
```

```
column_names on
delimiter ,
enclosures ""
encoding UTF8
row_terminator default
```

```
** UNLOAD Start ** at 2026.03.17-08.46.50
Export Separate Files to /Users/rg/DEMO
DATA TABLE SAMPLE_RESTAURANT
File Name: /Users/rg/DEMO/SAMPLE_RESTAURANT_DATA_TABLE.csv
Number of Rows Exported: 11
** UNLOAD End ** at 2026.03.17-08.46.51
SQL> |
```

| SAMPLE_RESTAURANT_DATA_TABLE.csv        |                                                                              |
|-----------------------------------------|------------------------------------------------------------------------------|
| SAMPLE_RESTAURANT_DATA_TABLE.csv > data |                                                                              |
| 1                                       | "ID","NAME","ADDRESS","RATING","LOGO_URL","HERO_URL","CATEGORY_ID","CREATED" |
| 2                                       | 1,"ByteStart Breakfast Nook","Address #123","4.0","restaurant/bytestart-br   |
| 3                                       | 2,"SQLicious Bistro","Address #724","3.2","restaurant/sqlicious-bistro.jpg"  |
| 4                                       | 3,"PL/SQLoely Diner","Address #335","5.0","restaurant/sqloely-diner.jpg"     |
| 5                                       | 4,"Oracool Cafe","Address #294","4.5","restaurant/oracool-cafe.jpg","resta   |
| 6                                       | 5,"APEXquisite Delights","Address #513","4.3","restaurant/apexquisite-deli   |
| 7                                       | 6,"SQLtastic Grill","Address #367","3.7","restaurant/sqltastic-grill.jpg","  |
| 8                                       | 7,"PL/SQL Paradise","Address #211","4.9","restaurant/sql-paradise.jpg","re   |
| 9                                       | 8,"APEXtraordinary Fare","Address #841","2.9","restaurant/apextraordinary-   |
| 10                                      | 9,"Oraculicious Cuisine","Address #751","4.7","restaurant/oraculicious-cui   |
| 11                                      | 10,"APEX Mex Grill","Address #571","5.0","restaurant/apex-mex-grill.jpg","   |
| 12                                      | 11,"SQL Sushi Bar","Address #382","4.7","restaurant/sql-sushi-grill.jpg","   |
| 13                                      |                                                                              |





- > set loadformat json
- > unload sample\_restaurant

```
SQL> set loadformat json
SQL> unload sample_restaurant

format json

** UNLOAD Start ** at 2026.03.17-08.51.04
Export Separate Files to /Users/rg/DEMO
DATA TABLE SAMPLE_RESTAURANT
File Name: /Users/rg/DEMO/SAMPLE_RESTAURANT_DATA_TABLE.json
Number of Rows Exported: 11
** UNLOAD End ** at 2026.03.17-08.51.04
SQL> 
```

```
{ } SAMPLE_RESTAURANT_DATA_TABLE.json X
{ } SAMPLE_RESTAURANT_DATA_TABLE.json > ...

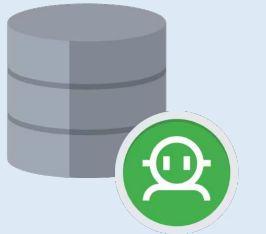
1  {"results":[{"columns":[{"name":"ID","type":"NUMBER"}, {"name":"NAME"
2  [
3  {"id":1,"name":"ByteStart Breakfast Nook","address":"Address #123",
4  ,{"id":2,"name":"SQLicious Bistro","address":"Address #724","rating"
5  ,{"id":3,"name":"PL\SQLoely Diner","address":"Address #335","rati
6  ,{"id":4,"name":"Oracool Cafe","address":"Address #294","rating":"4.
7  ,{"id":5,"name":"APEXquisite Delights","address":"Address #513","rat
8  ,{"id":6,"name":"SQLtastic Grill","address":"Address #367","rating":
9  ,{"id":7,"name":"PL\SQL Paradise","address":"Address #211","rating"
10 ,{"id":8,"name":"APEXtraordinary Fare","address":"Address #841","rat
11 ,{"id":9,"name":"Oraculicious Cuisine","address":"Address #751","rat
12 ,{"id":10,"name":"APEX Mex Grill","address":"Address #571","rating":
13 ,{"id":11,"name":"SQL Sushi Bar","address":"Address #382","rating":
14 ]}]}
```



# Import data using SQLcl



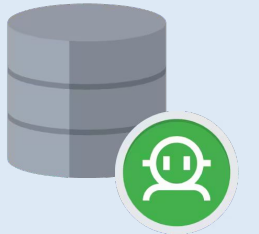
Scan to ask a question



- > **load** [table\_name] [file\_name] [options]
- > **load** conference sample\_restaurant\_data\_table.csv **NEW**

| SAMPLE_RESTAURANT_DATA_TABLE.csv        |                                                                        |
|-----------------------------------------|------------------------------------------------------------------------|
| SAMPLE_RESTAURANT_DATA_TABLE.csv > data |                                                                        |
| 1                                       | "ID","NAME","ADDRESS","RATING","LOGO_URL","HERO_URL","CATEGORY_ID","CR |
| 2                                       | 1,"ByteStart Breakfast Nook","Address #123","4.0","restaurant/bytest   |
| 3                                       | 2,"SQLicious Bistro","Address #724","3.2","restaurant/sqlicious-bist   |
| 4                                       | 3,"PL/SQLovely Diner","Address #335","5.0","restaurant/sqlovely-dine   |
| 5                                       | 4,"Oracool Cafe","Address #294","4.5","restaurant/oracool-cafe.jpg",   |
| 6                                       | 5,"APEXquisite Delights","Address #513","4.3","restaurant/apexquisit   |
| 7                                       | 6,"SQLtastic Grill","Address #367","3.7","restaurant/sqltastic-grill   |
| 8                                       | 7,"PL/SQL Paradise","Address #211","4.9","restaurant/sql-paradise.jp   |
| 9                                       | 8,"APEXtraordinary Fare","Address #841","2.9","restaurant/apextraord   |
| 10                                      | 9,"Oraculicious Cuisine","Address #751","4.7","restaurant/oraculicio   |
| 11                                      | 10,"APEX Mex Grill","Address #571","5.0","restaurant/apex-mex-grill.   |
| 12                                      | 11,"SQL Sushi Bar","Address #382","4.7","restaurant/sql-sushi-grill.   |





```
SQL> load conference sample_restaurant_data_table.csv NEW
```

```
csv
column_names on
delimiter ,
enclosures ""
double off
encoding UTF8
row_limit off
row_terminator default
skip_rows 0
skip_after_names
```

Create new table and load data into table DEMO.CONFERENCE

#INFO DATE format detected: DD-MON-RR

```
CREATE TABLE DEMO.CONFERENCE
```

```
(
  ID NUMBER(4),
  NAME VARCHAR2(128),
  ADDRESS VARCHAR2(26),
  RATING NUMBER(4, 1),
  LOGO_URL VARCHAR2(128),
  HERO_URL VARCHAR2(128),
  CATEGORY_ID NUMBER(3),
  CREATED DATE,
  CREATED_BY VARCHAR2(26),
  UPDATED DATE,
  UPDATED_BY VARCHAR2(26),
  DELIVERY_FEE NUMBER(3),
  DELIVERY_TIME NUMBER(4)
);
```

#INFO Table created

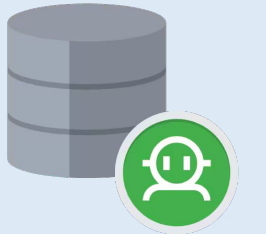
#INFO Number of rows processed: 11

#INFO Number of rows in error: 0

#INFO Last row processed in final committed batch: 11

SUCCESS: Processed without errors





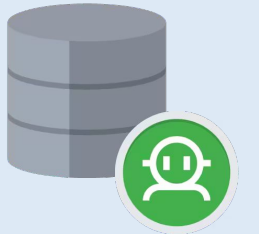
> **load** conference sample\_restaurant\_data\_table.csv

```
SQL> load conference sample_restaurant_data_table.csv
```

```
csv
column_names on
delimiter ,
enclosures ""
double off
encoding UTF8
row_limit off
row_terminator default
skip_rows 0
skip_after_names
```

```
Load data into table DEMO.CONFERENCE
```

```
#INFO Number of rows processed: 11
#INFO Number of rows in error: 0
#INFO Last row processed in final committed batch: 11
SUCCESS: Processed without errors
SQL> |
```



> **help** set loadformat

```
SQL> help set loadformat
```

```
SET LOADFORMAT [ default|csv|delimited|html|insert|json|json-formatted|loader|t2|xml|xls|xlsx] [options...]
```

|                |                                                                     |
|----------------|---------------------------------------------------------------------|
| default        | : Load format properties return to default values                   |
| csv            | : comma separated values                                            |
| delimited      | : (CSV synonym) delimited format, comma separated values by default |
| html           | : UNLOAD only, Hypertext Markup Language                            |
| insert         | : UNLOAD only, SQL insert statements                                |
| json           | : UNLOAD only, Java Script Object Notation                          |
| json-formatted | : UNLOAD only, "pretty" formatted JSON                              |
| loader         | : UNLOAD only, Oracle SQLLoader format                              |
| t2             | : UNLOAD only, T2 Metrics                                           |
| xml            | : UNLOAD only, Extensible Markup Language                           |
| xls            | : Excel 97- Excel 2003 Workbook format                              |
| xlsx           | : Excel format for version 2007 or higher                           |

# SQLcl Data Pump



Scan to ask a question





# Export full schema from DEV [DEMO] [on-premise]

> dp export

```
SQL> dp export
Initiating DATA PUMP
Database Time Zone: VERSION:44 CON_ID:0
Log Location: DATA_PUMP_DIR:ESQL_770.LOG
Starting "DEMO"."ESQL_770":
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROCACT_SCHEMA/LOGREP
Processing object type SCHEMA_EXPORT/PACKAGE/PACKAGE_BODY
Processing object type SCHEMA_EXPORT/TABLE/INDEX/STATISTICS/INDEX_STATISTICS
Processing object type SCHEMA_EXPORT/TABLE/STATISTICS/TABLE_STATISTICS
Processing object type SCHEMA_EXPORT/TABLE/TABLE_DATA
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROCOBJ/SCHEDULER
Processing object type SCHEMA_EXPORT/SEQUENCE/SEQUENCE
Processing object type SCHEMA_EXPORT/TABLE/COMMENT
Processing object type SCHEMA_EXPORT/TABLE/TABLE
Processing object type SCHEMA_EXPORT/TABLE/IDENTITY_COLUMN
Processing object type SCHEMA_EXPORT/PACKAGE/PACKAGE_SPEC
Processing object type SCHEMA_EXPORT/FUNCTION/FUNCTION
Processing object type SCHEMA_EXPORT/PROCEDURE/PROCEDURE
Processing object type SCHEMA_EXPORT/PACKAGE/COMPILE_PACKAGE/PACKAGE_SPEC/ALTER_PACKAGE_SPEC
Processing object type SCHEMA_EXPORT/FUNCTION/ALTER_FUNCTION
Processing object type SCHEMA_EXPORT/PROCEDURE/ALTER_PROCEDURE
Processing object type SCHEMA_EXPORT/TABLE/INDEX/INDEX
Processing object type SCHEMA_EXPORT/VIEW/VIEW
Processing object type SCHEMA_EXPORT/TABLE/CONSTRAINT/CONSTRAINT
Processing object type SCHEMA_EXPORT/TABLE/TRIGGER
Processing object type SCHEMA_EXPORT/TABLE/CONSTRAINT/REF_CONSTRAINT
. . exported "DEMO"."ESQL_500"          2.4 MB      2218 rows
. . exported "DEMO"."ESQL_498"          2.2 MB      2196 rows
. . exported "DEMO"."ESQL_406"          2.1 MB      2173 rows
```

<<time>>

```
. . exported "DEMO"."EBA_DEMO_DG_PRODUCTS"      9.3 KB      46 rows
. . exported "DEMO"."EBA_DEMO_MD_COMMENTS"      9.2 KB      11 rows
. . exported "DEMO"."EBA_DEMO_DG_EMP"           9 KB       14 rows
. . exported "DEMO"."EBA_DEMO_TREE_EMP"         9 KB       14 rows
. . exported "DEMO"."EBA_UT_CHART_PROJECTS"     8.7 KB     14 rows
. . exported "DEMO"."SAMPLE_RESTAURANT_CATEGORIES" 8.5 KB     10 rows
. . exported "DEMO"."SAMPLE_RESTAURANT_ORDER_ITEMS" 8.1 KB      8 rows
. . exported "DEMO"."EBA_DEMO_MD_STATUS"        8 KB        3 rows
. . exported "DEMO"."EBA_DEMO_SEARCH_CUSTOMERS"  6.9 KB     20 rows
. . exported "DEMO"."SAMPLE_RESTAURANT_ORDERS"   6.6 KB      5 rows
. . exported "DEMO"."EBA_DEMO_IG_DEPT"          6.5 KB      4 rows
. . exported "DEMO"."EBA_DEMO_DG_DEPT"          6.1 KB      4 rows
. . exported "DEMO"."EBA_DEMO_TREE_DEPT"        6.1 KB      4 rows
. . exported "DEMO"."EBA_DEMO_SEARCH_WEB_CREDS"  0 KB        0 rows
. . exported "DEMO"."EBA_VECTOR_AI_HELPER"       0 KB        0 rows
. . exported "DEMO"."EBA_VECTOR_DATATABLE"       0 KB        0 rows
ORA-39173: Encrypted data has been stored unencrypted in dump file set.
Master table "DEMO"."ESQL_770" successfully loaded/unloaded
*****
Dump file set for DEMO.ESQL_770 is:
/u03/dbfs/4B003B659121328EE063BB59000ACD83/data/dpdump/ESQL_770.DMP
Job "DEMO"."ESQL_770" successfully completed at Tue Mar 17 16:47:19 2026 elapsed 0 00:01:15
DataPump Operation Status 'COMPLETED'
```

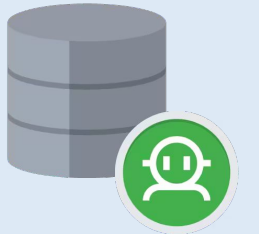




# Export full schema from DEV [DEMO] [OCI]

```
> datapump EXPORT -schemas DEMO -directory DATA_PUMP_DIR -credential
DEMO_OBJ_STORAGE -dumpuri MY_BUCKET_URL/demo_schema.dmp
```

```
. . exported "DEMO"."EBA_DEMO_MD_MILESTONES"          10.8 KB      12 rows
. . exported "DEMO"."EBA_DEMO_MD_TASK_LINKS"           11 KB         7 rows
. . exported "DEMO"."EBA_DEMO_SEARCH_ORDERS"          10.1 KB     105 rows
. . exported "DEMO"."EBA_DEMO_MD_PROJECTS"            10.1 KB         5 rows
. . exported "DEMO"."EBA_DEMO_DG_PRODUCTS"             9.3 KB        46 rows
. . exported "DEMO"."EBA_DEMO_MD_COMMENTS"             9.2 KB        11 rows
. . exported "DEMO"."EBA_DEMO_DG_EMP"                  9 KB          14 rows
. . exported "DEMO"."EBA_DEMO_TREE_EMP"                9 KB          14 rows
. . exported "DEMO"."EBA_UT_CHART_PROJECTS"            8.7 KB        14 rows
. . exported "DEMO"."SAMPLE_RESTAURANT_CATEGORIES"      8.5 KB        10 rows
. . exported "DEMO"."SAMPLE_RESTAURANT_ORDER_ITEMS"    8.1 KB         8 rows
. . exported "DEMO"."EBA_DEMO_MD_STATUS"               8 KB           3 rows
. . exported "DEMO"."EBA_DEMO_SEARCH_CUSTOMERS"        6.9 KB        20 rows
. . exported "DEMO"."SAMPLE_RESTAURANT_ORDERS"         6.6 KB         5 rows
. . exported "DEMO"."EBA_DEMO_IG_DEPT"                 6.5 KB         4 rows
. . exported "DEMO"."EBA_DEMO_DG_DEPT"                 6.1 KB         4 rows
. . exported "DEMO"."EBA_DEMO_TREE_DEPT"               6.1 KB         4 rows
. . exported "DEMO"."EBA_DEMO_SEARCH_WEB_CREDS"         0 KB           0 rows
. . exported "DEMO"."EBA_VECTOR_AI_HELPER"              0 KB           0 rows
. . exported "DEMO"."EBA_VECTOR_DATATABLE"              0 KB           0 rows
ORA-39173: Encrypted data has been stored unencrypted in dump file set.
Master table "DEMO"."ESQL_772" successfully loaded/unloaded
*****
Dump file set for DEMO.ESQL 772 is:
https://objectstorage[redacted]/demo_schema.dmp
Job "DEMO"."ESQL_772" successfully completed at Tue Mar 17 17:05:51 2026 elapsed 0 00:00:57
DataPump Operation Status 'COMPLETED'
```



## Import full schema [DEMO] into TEST

> sql -name test\_demo

> datapump IMPORT -schemas DEMO -directory DATA\_PUMP\_DIR -credential  
DEMO\_OBJ\_STORAGE -dumpuri MY\_BUCKET\_URL/o/demo\_schema.dmp

```
SQL> DATAPUMP IMPORT -schemas DEMO -directory DATA_PUMP_DIR -credential DEMO_OBJ_STORAGE -dumpuri /bucket-data-pump-demo/o/demo_schema.dmp
Initiating DATA PUMP
Database Time Zone: VERSION:44 CON_ID:0
Log Location: DATA_PUMP_DIR:ISQL_420.LOG
Master table "DEMO"."ISQL_420" successfully loaded/unloaded
Starting "DEMO"."ISQL_420":
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROCACT_SCHEMA/LOGREP
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROCOBJ/SCHEDULER
ORA-39083: Object type SCHEDULER:PROCOBJ failed to create with error:
ORA-20022: Credential "DEMO"."DEMO_OBJ_STORAGE" already exists
Processing object type SCHEMA_EXPORT/SEQUENCE/SEQUENCE
Processing object type SCHEMA_EXPORT/TABLE/TABLE
Processing object type SCHEMA_EXPORT/TABLE/TABLE_DATA
. . imported "DEMO"."EBA_DEMO_SEARCH_WEB_CREDS"          0 KB          0 rows
. . imported "DEMO"."EBA_VECTOR_DATATABLE"              0 KB          0 rows
. . imported "DEMO"."EBA_DEMO_SEARCH_CUSTOMERS"         6.9 KB         20 rows
. . imported "DEMO"."EBA_DEMO_SEARCH_ORDERS"            10.1 KB        105 rows
. . imported "DEMO"."EBA_DEMO_DG_DEPT"                  6.1 KB          4 rows
. . imported "DEMO"."SAMPLE_RESTAURANT_ITEMS"           12.8 KB         24 rows
. . imported "DEMO"."EBA_UT_DEMO_CARDS"                 19.4 KB         18 rows
. . imported "DEMO"."EBA_DEMO_SEARCH_PRODUCTS"          45.1 KB         46 rows
. . imported "DEMO"."EBA_DEMO_MD_TASKS"                 19.1 KB         45 rows
. . imported "DEMO"."SAMPLE_RESTAURANT"                 12.1 KB         11 rows
. . imported "DEMO"."EBA_DEMO_DG_EMP"                   9 KB           14 rows
. . imported "DEMO"."EBA_DEMO_MD_TASK_TODOS"            11.2 KB          9 rows
. . imported "DEMO"."EBA_DEMO_IG_DEPT"                   6.5 KB          4 rows
. . imported "DEMO"."EBA_DEMO_MD_PROJECTS"              10.1 KB          5 rows
. . imported "DEMO"."EBA_DEMO_DG_CUSTOMERS"              25 KB        392 rows
. . imported "DEMO"."EBA_DEMO_IG_EMP"                   12.6 KB         15 rows
. . imported "DEMO"."EBA_DEMO_TREE_STOCKS"              11.6 KB         11 rows
. . imported "DEMO"."ESQL_770"                          2.6 MB       2240 rows
. . imported "DEMO"."EBA_DEMO_SEARCH_STORES"            11.6 KB         23 rows
. . imported "DEMO"."ESQL_408"                          2.2 MB       2106 rows
```

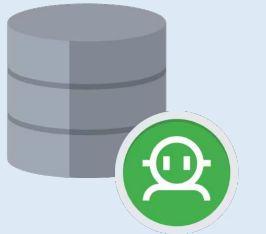




# Import full schema [DEMO] into TEST

```
> sql -name test_demo
> datapump IMPORT -schemas DEMO -directory DATA_PUMP_DIR -credential
DEMO_OBJ_STORAGE -dumpuri MY_BUCKET_URL/o/demo_schema.dmp
```

```
. . imported "DEMO"."SAMPLE_RESTAURANT_ORDERS"          6.6 KB      5 rows
. . imported "DEMO"."SAMPLE_RESTAURANT_ORDER_ITEMS"     8.1 KB      8 rows
. . imported "DEMO"."EBA_DEMO_TREE_DEPT"                6.1 KB      4 rows
. . imported "DEMO"."CONFERENCE"                       13.8 KB     22 rows
. . imported "DEMO"."ESQL_500"                          2.4 MB    2218 rows
. . imported "DEMO"."EBA_DEMO_MD_COMMENTS"              9.2 KB     11 rows
. . imported "DEMO"."SAMPLE_RESTAURANT_NEW"            12.2 KB     11 rows
. . imported "DEMO"."EBA_DEMO_DG_PRODUCTS"              9.3 KB     46 rows
. . imported "DEMO"."EBA_DEMO_MD_MILESTONES"            10.8 KB     12 rows
. . imported "DEMO"."EBA_VECTOR_MOVEDATA"               10 KB      10 rows
. . imported "DEMO"."EBA_DEMO_MD_TEAM_MEMBERS"          11 KB      12 rows
. . imported "DEMO"."SAMPLE_RESTAURANT_CATEGORIES"       8.5 KB     10 rows
. . imported "DEMO"."EBA_DEMO_TREE_PROJ_FILES"           51 KB       4 rows
. . imported "DEMO"."EBA_UT_MAP_AIRPORTS"               20.7 KB    45 rows
Processing object type SCHEMA_EXPORT/TABLE/COMMENT
Processing object type SCHEMA_EXPORT/TABLE/IDENTITY_COLUMN
Processing object type SCHEMA_EXPORT/PACKAGE/PACKAGE_SPEC
Processing object type SCHEMA_EXPORT/FUNCTION/FUNCTION
Processing object type SCHEMA_EXPORT/PROCEDURE/PROCEDURE
Processing object type SCHEMA_EXPORT/PACKAGE/COMPILE_PACKAGE/PACKAGE_SPEC/ALTER_PACKAGE_SPEC
Processing object type SCHEMA_EXPORT/FUNCTION/ALTER_FUNCTION
Processing object type SCHEMA_EXPORT/PROCEDURE/ALTER_PROCEDURE
Processing object type SCHEMA_EXPORT/VIEW/VIEW
Processing object type SCHEMA_EXPORT/PACKAGE/PACKAGE_BODY
Processing object type SCHEMA_EXPORT/TABLE/INDEX/INDEX
Processing object type SCHEMA_EXPORT/TABLE/CONSTRAINT/CONSTRAINT
Processing object type SCHEMA_EXPORT/TABLE/INDEX/STATISTICS/INDEX_STATISTICS
Processing object type SCHEMA_EXPORT/TABLE/CONSTRAINT/REF_CONSTRAINT
Processing object type SCHEMA_EXPORT/TABLE/TRIGGER
Processing object type SCHEMA_EXPORT/TABLE/STATISTICS/TABLE_STATISTICS
Processing object type SCHEMA_EXPORT/POST_SCHEMA/PROCACT_SCHEMA/LBAC_EXP
Job "DEMO"."ISQL_420" completed with 1 error(s) at Tue Mar 17 17:25:42 2026 elapsed 0 00:00:55
DataPump Operation Status 'COMPLETED'
```



# Do you want to know more?

## > **help** datapump

Oracle Data Pump enables high-speed movement of data and metadata from one database to another. The datapump command uses the extension of the DBMS\_DATAPUMP package. The subcommands **export** and **import** are available within the **datapump** feature.

### Subcommands:

#### **EXPORT**

Export schema or list of schemas with the datapump utility using DBMS\_DATAPUMP package.

#### Usage:

```
export|ex {OPTIONS}
```

#### Options:

```
-compression|-cm <compression> {all|data_only|metadata_only|none}
```

Default: As specified on set datapump or METADATA\_ONLY

Indicates if compression should be done on user data and metadata.

ALL : Compress user data and metadata.

DATA\_ONLY : Compress only user data.

METADATA\_ONLY : Compress only metadata.

NONE : Do not compress user data or metadata.

```
-copycloud|-cc <copycloud>
```

Default: As specified on set datapump or FALSE

[TRUE] : Copy the dumpfile between database directory and Oracle Object Store

On export, copy the dumpfile from the database directory to Oracle Object Store after the datapump job completes.

On import, copy the dumpfile from Oracle Object Store to the database directory before the datapump job starts

OCI PROFILE must be set using OCI command or CLOUDSTORAGE command.

Cloud storage uri must be set using the CLOUDSTORAGE command, SET DATAPUMP command or DATAPUMP command.



pretius

# SHOW LEVEL & SCRIPTS

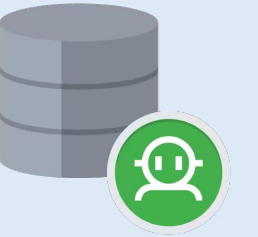
(new commands since May 2026)



Scan to ask a question

 **ASCEND**  
2026





## > show level

Displays the current script nesting depth and current script path

## > show scripts

Displays the current script call stack



EXPLORER

> OPEN EDITORS

DEMO\_SQLCL

- > apex
- > show\_level\_and\_scripts
  - includes
    - level1
      - level2
        - level3
          - level4
            - level5
              - level5.sql
              - level4.sql
              - level3.sql
              - level2.sql
              - level1.sql
              - demo.md
              - main.sql
              - README.md

main.sql U X

show\_level\_and\_scripts > main.sql > ...

```
1 prompt === main.sql start ===
2 show scripts
3 show level
4 select 'my MAIN level query' as script_name from dual;
5 @@includes/level1/level1.sql
```

level1.sql U X

show\_level\_and\_scripts > includes > level1 > level1.sql > ...

```
1 select 'my level1 query' as script_name from dual;
2 @@level2/level2.sql
```

level5.sql U X

show\_level\_and\_scripts > includes > level1 > level2 > level3 > level4 > level5 > level5.sql > ...

```
1 show scripts
2 show level
3 select 'my level5 query' as script_name from dual;
```



show\_level\_and\_scripts

includes

level1

level2

level3

level4

level5

level5.sql

level4.sql

level3.sql

level2.sql

level1.sql

demo.md

main.sql

README.md

PROBLEMS

OUTPUT

TERMINAL

CHAT

bash - show\_level\_and\_scripts

DEMO\_RAFAŁ@demo\_rafał > |





show\_level\_and\_scripts

includes

level1

level2

level3

level4

level5

level5.sql

level4.sql

level3.sql

level2.sql

level1.sql

demo.md

main.sql

README.md

PROBLEMS

OUTPUT

TERMINAL

CHAT

bash - show\_level\_and\_scripts

DEMO RAFAL@demo rafal > @main.sql;

=== main.sql start ===

scripts

└─ main.sql

depth: 2

current: /Users/rg/git/sandbox/demo\_sqlcl/show\_level\_and\_scripts/main.sql

SCRIPT\_NAME

my MAIN level query

SCRIPT\_NAME

my level1 query

SCRIPT\_NAME

my level2 query

SCRIPT\_NAME

my level3 query

SCRIPT\_NAME

my level4 query

scripts

└─ main.sql

└─ includes/level1/level1.sql

└─ level2/level2.sql

└─ level3/level3.sql

└─ level4/level4.sql

└─ level5/level5.sql

depth: 7

current: /Users/rg/git/sandbox/demo\_sqlcl/show\_level\_and\_scripts/includes/level1/level2/level3/level4/level5/level5.sql

SCRIPT\_NAME

my level5 query

# APEX [subcommand]



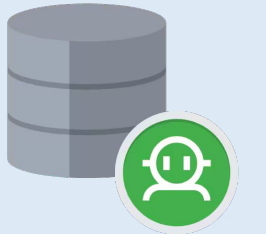
Scan to ask a question



## APEX Applications subcommands

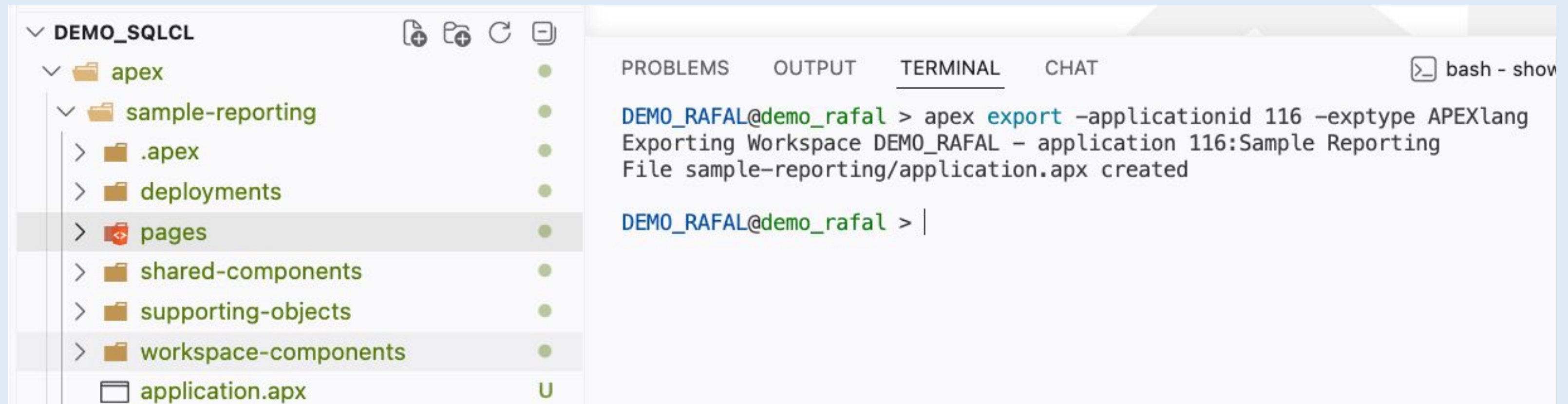
- > apex [subcommand]
- > export (export-all-applications)
- > export-all-workspaces (wkspc only, no apps, structure only)
- > export-components
- > list
- > log (usage reports)
- > version
- > import (APEXlang support since May 2026)



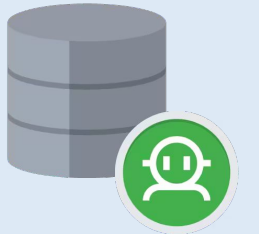


# Export APEX application

> apex export -applicationid 116 -exptype APEXLANG-split -dir apex



\* Optionally export all apps with **export-all-applications -workspaceid 123456789**



## Validate exported app

> apex **validate** -input sample-reporting -workspace DEMO\_RAFAL

```
DEMO_RAFAL@demo_apexlang > apex validate -input sample-reporting -workspace DEMO_RAFAL  
Validation successful.
```

```
DEMO_RAFAL@demo_apexlang > |
```

```
DEMO_RAFAL@demo_apexlang > apex validate -input sample-reporting -workspace DEMO_RAFAL
```

APEXLang Compile Errors:

File: pages/p00001-interactive-report.apx

Line: 268

Column: 8

Type: DUPLICATED\_COMPONENT

Error: Duplicated component within parent scope. For properties:  
advanced.endUserAlias: L

File: pages/p00001-interactive-report.apx

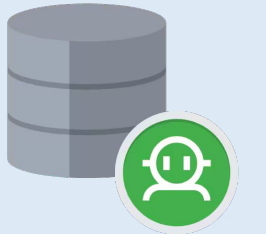
Line: 343

Column: 8

Type: DUPLICATED\_COMPONENT

Error: Duplicated component within parent scope. For properties:  
advanced.endUserAlias: L





## Other APEX export types

> help apex export

```
-exptype|-exty <exptype> {APEXLANG|APPLICATION_SOURCE|CHECKSUM-SH1|CHECKSUM-SH256|EMBEDDED_CODE|READABLE_JSON|READABLE_YAML|SQL,...} (APPLICATION_SOURCE)
```

Comma delimited list of export types to perform:

- \* APPLICATION\_SOURCE: Export an APEX application using other parameters passed.
- \* APEXLANG: Export a readable version of the application meta-data in APEXlang format.
- \* SQL: Export an APEX application in to SQL format.
- \* EMBEDDED\_CODE: Export code such as SQL, PL/SQL and Javascript. APEX ignores all other options when EMBEDDED\_CODE is selected.
- \* CHECKSUM-SH1: Export a SHA1 checksum that is independent of IDs and can be compared across instances and workspaces.
- \* CHECKSUM-SH256: Export a SHA-256 checksum that is independent of IDs and can be compared across instances and workspaces.
- \* READABLE\_JSON: Export a readable version of the application meta-data in JSON format.
- \* READABLE\_YAML: Export a readable version of the application meta-data in YAML format.

On APEX version 26.1 and after readable formats will be exported as APEXlang.



# List - returns all APEX applications

> apex list

```
SQL> apex list
```

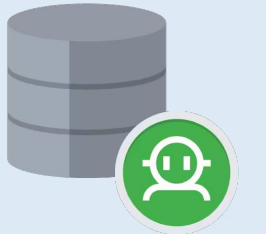
| WORKSPACE_ID      | WORKSPACE | APPLICATION_ID | APPLICATION_NAME               | BUILD_STATUS    | LAST_UPDATED_ON | LAST_UPDATED_BY |
|-------------------|-----------|----------------|--------------------------------|-----------------|-----------------|-----------------|
| 11962392022442402 | DEMO      | 103            | APEXToGo                       | Run and Develop |                 |                 |
| 11962392022442402 | DEMO      | 104            | Sample Interactive Grids       | Run and Develop |                 |                 |
| 11962392022442402 | DEMO      | 105            | Sample Trees                   | Run and Develop |                 |                 |
| 11962392022442402 | DEMO      | 106            | Sample Document Generator      | Run and Develop |                 |                 |
| 11962392022442402 | DEMO      | 107            | Universal Theme 24.2 Reference | Run and Develop |                 |                 |
| 11962392022442402 | DEMO      | 109            | Sample Vector Search           | Run and Develop |                 |                 |
| 11962392022442402 | DEMO      | 110            | Sample Master Detail           | Run and Develop |                 |                 |
| 11962392022442402 | DEMO      | 111            | Sample Application Search      | Run and Develop |                 |                 |

# Version & log - current APEX version and apps usage

```
> apex version
```

```
> apex log [APEX_WORKSPACE_LOG_SUMMARY]
```

[illegible]



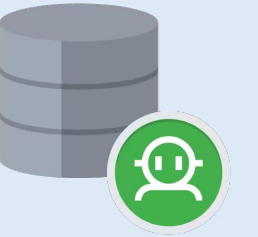
## Export components

> apex **export-components** -api 103 -expcomponents PAGE:1

```
DEMO@dev_demo > apex export-components -api 103 -expcomponents PAGE:1
Exporting Workspace DEMO - application 103:Sample Reporting
File f103_page_1.sql created

DEMO@dev_demo > |
```





> **help** apex export

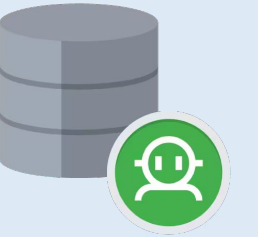
```
DEMO_RAFAŁ@demo_rafał > |
```



# SQLcl Project (CI/CD)

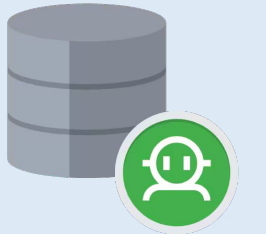


Scan to ask a question



- “Project ” is an CI/CD extension in SQLcl
- Introduced in 2nd half of 2024
- Contains several sub-commands to help developers manage database & APEX changes
- It uses **Liquibase** & **GIT** in a simple & developer-friendly way





## Initialize a new project (GIT repo required)

- > sql -name **dev\_demo**
- > project **init** -name DEMO -schemas DEMO

```
DEMO@dev_demo > project init -name DEMO -schemas DEMO
```

### PROJECT DETAILS

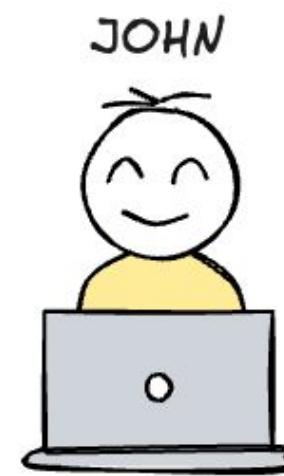
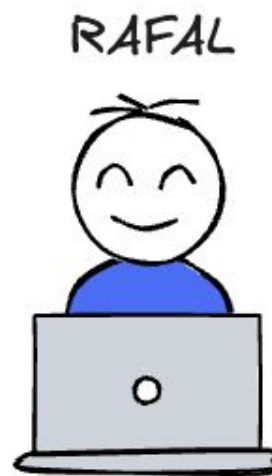
```
-----  
Project name:    DEMO  
Schema(s):      DEMO  
Directory:       /Users/rg/DEMO  
Connection name: dev_demo  
Project root:    DEMO  
Your project has been successfully created  
DEMO@dev_demo >
```

A file explorer window showing the contents of the DEMO project. The project folder is expanded, revealing subfolders .dbtools, dist, and src. It also shows files .gitignore and README.md. The .gitignore and README.md files have a 'U' icon next to them, indicating they are untracked.

```
▼ DEMO  
  > .dbtools  
  > dist  
  > src  
  .gitignore U  
  README.md U
```



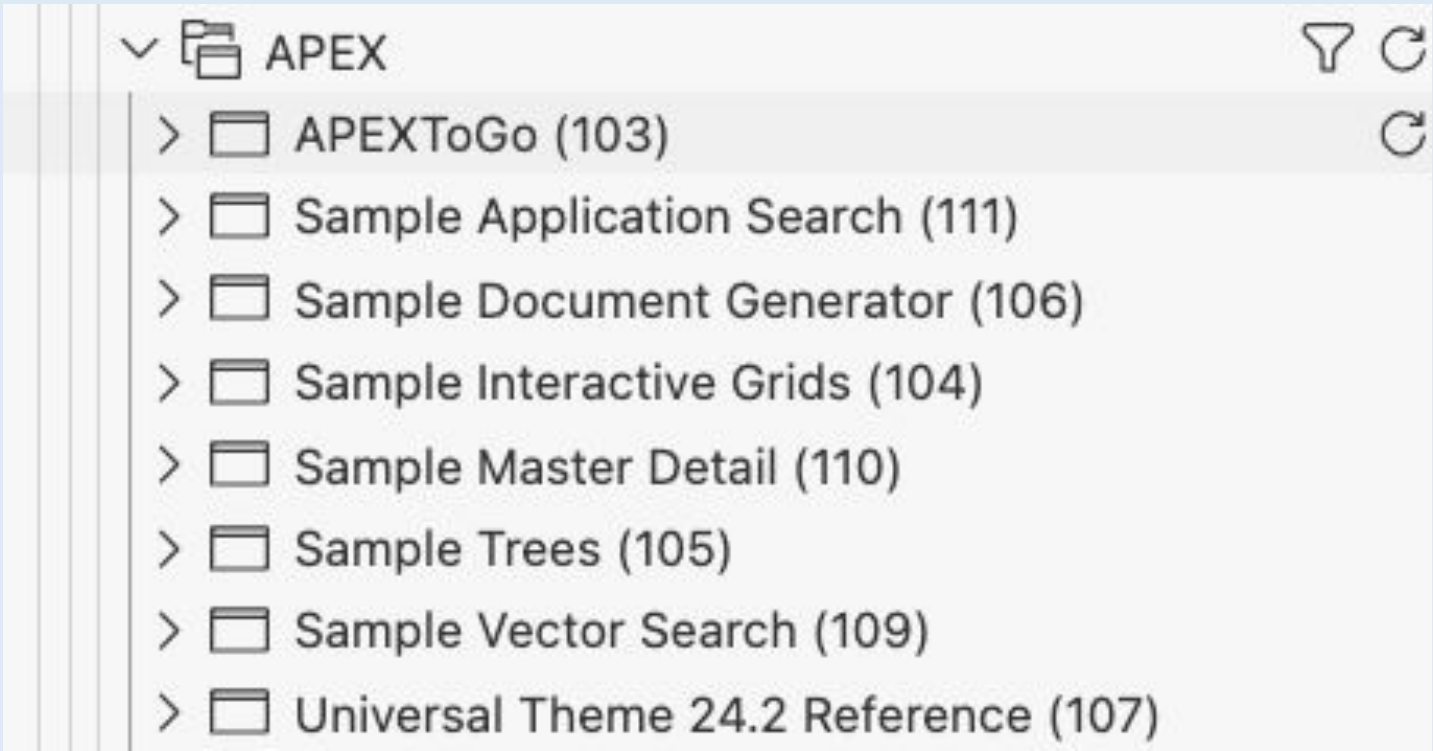
Make some Database Objects & APEX changes





# Export ALL database & APEX objects

| COUNT(*) | OBJECT_TYPE  |
|----------|--------------|
| 142      | INDEX        |
| 51       | TABLE        |
| 30       | LOB          |
| 1        | CREDENTIAL   |
| 23       | TRIGGER      |
| 6        | PACKAGE      |
| 7        | VIEW         |
| 2        | FUNCTION     |
| 6        | PACKAGE BODY |
| 14       | SEQUENCE     |
| 2        | PROCEDURE    |



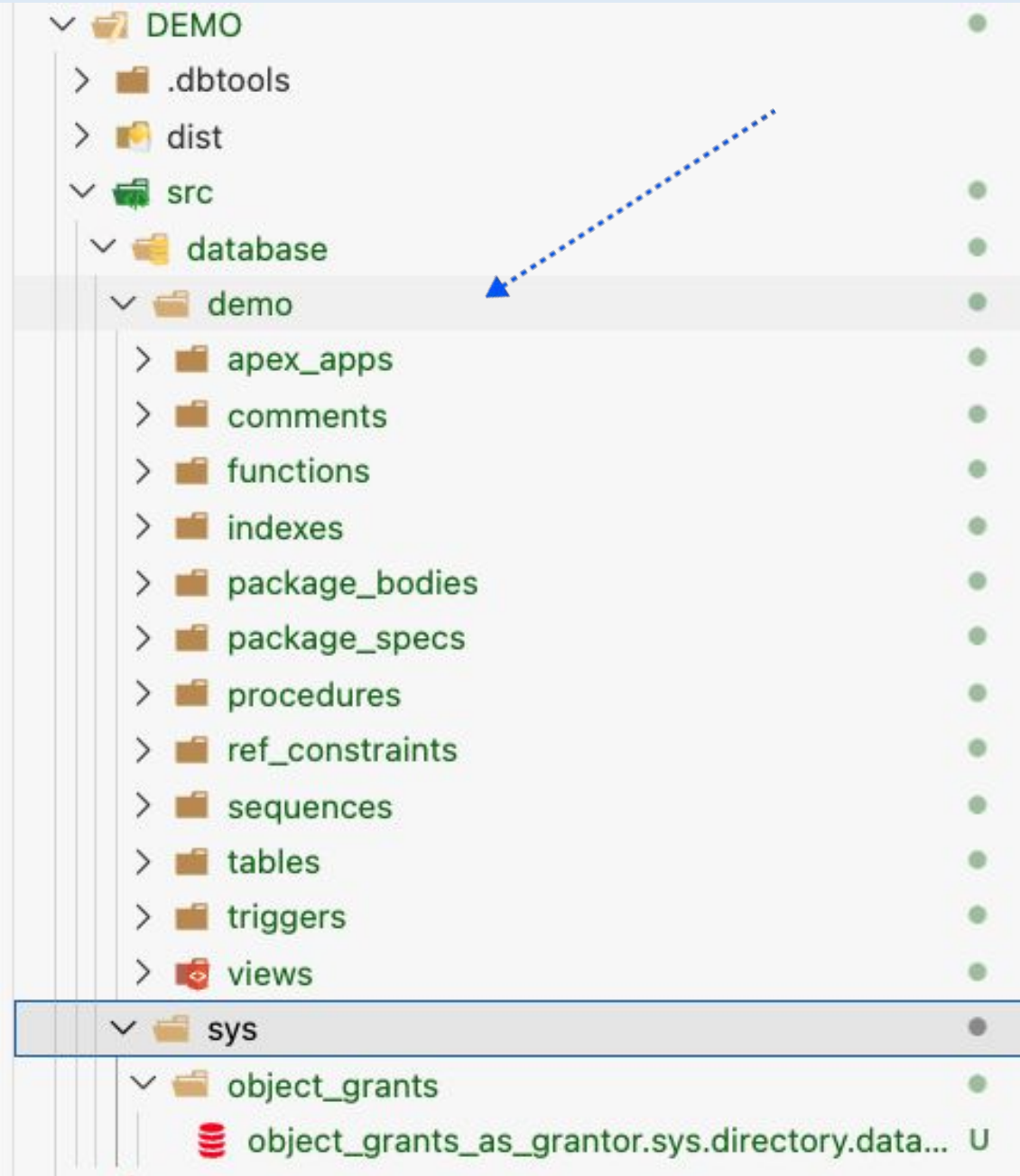




# Export ALL database & APEX objects

> project export

```
DEMO@dev_demo > project export
The current connection DEMO will
*** TABLES ***
*** OBJECT_GRANTS ***
*** APEX_APPLICATIONS ***
*** VIEWS ***
*** FUNCTIONS ***
*** PACKAGE_SPECS ***
*** PACKAGE_BODIES ***
*** PROCEDURES ***
*** SEQUENCES ***
*** INDEXES ***
*** TRIGGERS ***
*** REF_CONSTRAINTS ***
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
-----
APEX_APPLICATION      8
COMMENT                5
FUNCTION                2
GRANT                  2
INDEX                 61
PACKAGE_BODY           6
PACKAGE_SPEC           6
PROCEDURE              2
REF_CONSTRAINT        30
SEQUENCE               3
TABLE                 51
TRIGGER               23
VIEW                   7
-----
Exported 206 objects
Elapsed 208 sec
```





## Project Stage

- Compares the current branch (**dev**) with the target branch (**test**) to create Liquibase changelogs that apply the necessary database updates.

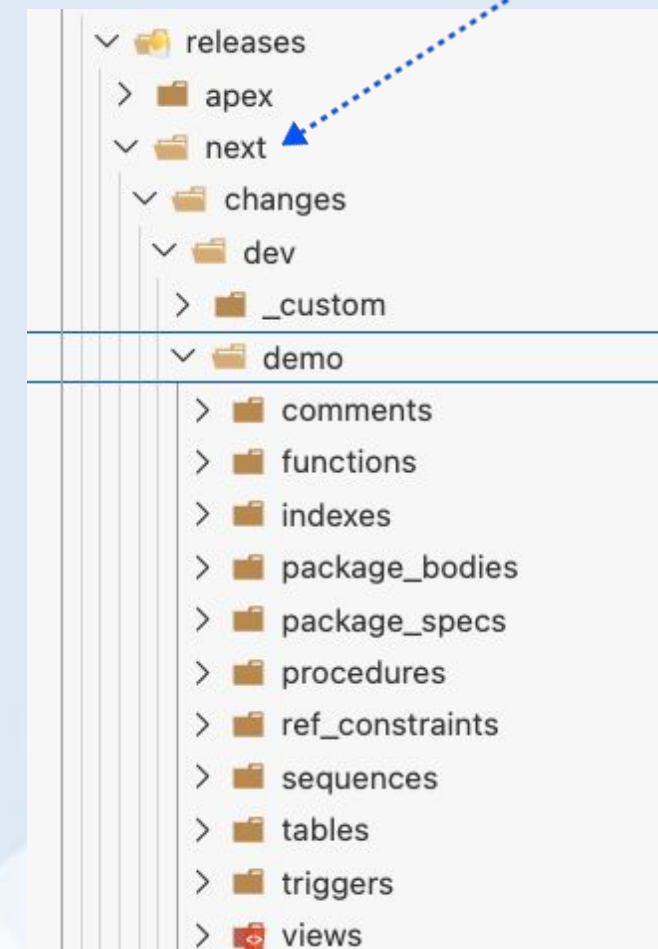
> project **stage** -branch-name test

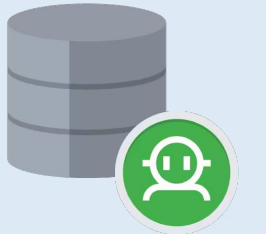
```
DEMO@dev_demo > project stage -branch-name test
The current connection DEMO will be used for all operations
```

Stage is Comparing:

|            |                 |
|------------|-----------------|
| Old Branch | refs/heads/test |
| New Branch | refs/heads/dev  |

Stage processing completed, please review and commit your changes to repository



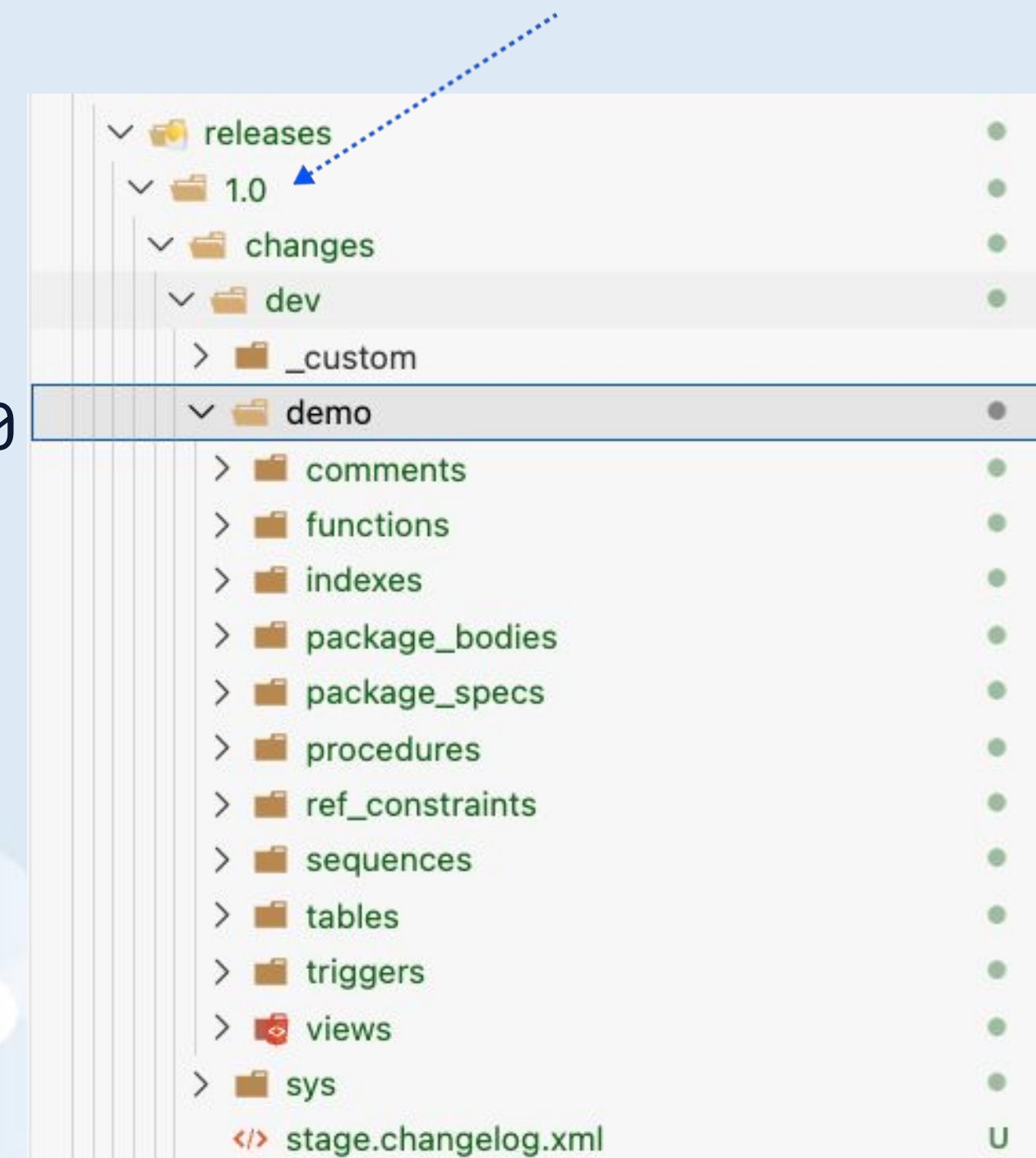


## Project Release

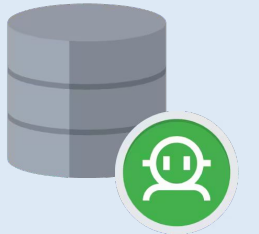
- Transitions current work set into a released state and initiates a fresh set of work
- Once the code is moved into a release, it must not be altered.

> project **release** -version 1.0

```
DEMO@dev_demo > project release -version 1.0
Process completed successfully
DEMO@dev_demo > |
```







Liquibase syntax and files are automatically created:

eba\_demo\_dg\_orders.sql U X

DEMO > dist > releases > 1.0 > changes > dev > demo > tables > eba\_demo\_dg\_orders.sql > ...

```
1  -- liquibase formatted sql
2  -- changeset DEMO:1773775410323 stripComments:false logicalFilePath:dev/demo/tables/eba_demo_dg_orders.sql
3  -- sqlcl_snapshot src/database/demo/tables/eba_demo_dg_orders.sql:null:dc1ca737d177f7fef60aaa0dffff9f9bf1b8458e:create
4
5  create table demo.eba_demo_dg_orders (
6      order_id      number generated by default on null as identity not null enable,
7      order_datetime timestamp(6) not null enable,
8      customer_id    number not null enable,
9      order_status   varchar2(10 byte) not null enable
10 );
```

stage.changelog.xml U X

DEMO > dist > releases > 1.0 > changes > dev > stage.changelog.xml

```
1  <?xml version="1.0" encoding="UTF-8"?>
2  <databaseChangelog xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3      xmlns="http://www.liquibase.org/xml/ns/dbchangelog"
4      xsi:schemaLocation="http://www.liquibase.org/xml/ns/dbchangelog
5          http://www.liquibase.org/xml/ns/dbchangelog/dbchangelog-4.3.xsd">
6      <include file="sys/object_grants/object_grants_as_grantor.sys.directory.data_pump_dir.sql" relativeToChangelogFile="true"/>
7      <include file="demo/sequences/eba_demo_ig_dept_seq.sql" relativeToChangelogFile="true"/>
8      <include file="demo/sequences/eba_demo_ig_emp_seq.sql" relativeToChangelogFile="true"/>
9      <include file="demo/sequences/eba_demo_ig_people_seq.sql" relativeToChangelogFile="true"/>
10     <include file="demo/functions/varchar2_to_blob.sql" relativeToChangelogFile="true"/>
11     <include file="demo/package_specs/eba_demo_ig_text_pkg.sql" relativeToChangelogFile="true"/>
12     <include file="demo/package_specs/eba_demo_md_data_pkg.sql" relativeToChangelogFile="true"/>
13     <include file="demo/package_specs/eba_demo_search_data_pkg.sql" relativeToChangelogFile="true"/>
14     <include file="demo/package_specs/eba_demo_search_text_pkg.sql" relativeToChangelogFile="true"/>
15     <include file="demo/package_specs/eba_sample_vector_search_pkg.sql" relativeToChangelogFile="true"/>
16     <include file="demo/tables/eba_demo_dg_dept.sql" relativeToChangelogFile="true"/>
17     <include file="demo/tables/eba_demo_dg_emp.sql" relativeToChangelogFile="true"/>
18     <include file="demo/tables/eba_demo_ig_dept.sql" relativeToChangelogFile="true"/>
```

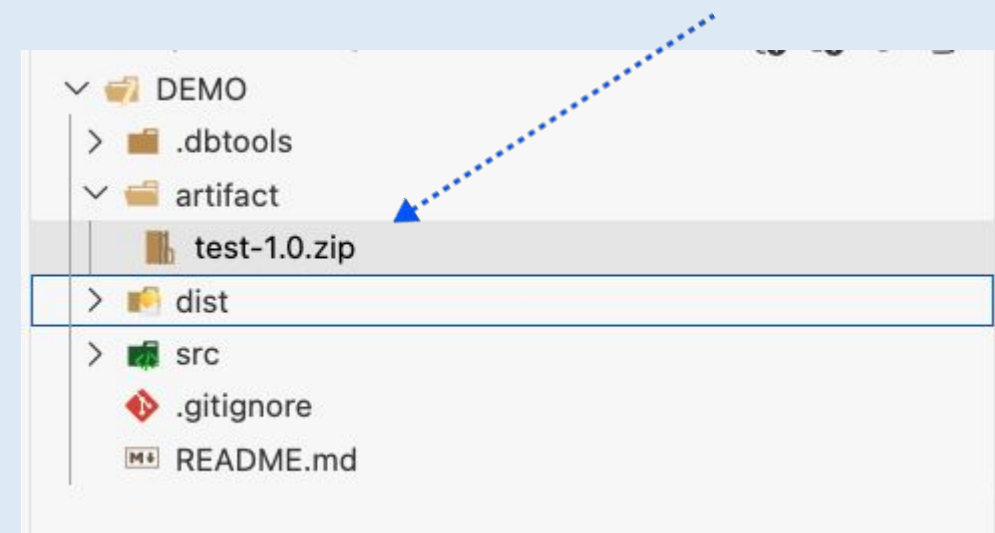


## Project Gen-Artifact

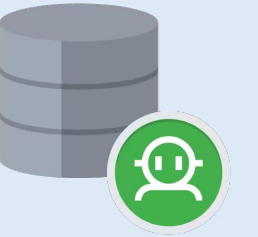
- Generates an artifact representing the current state of your project.

> project **gen-artifact** -name test -version 1.0

```
DEMO@dev_demo > project gen-artifact -name test -version 1.0  
Your artifact has been generated test-1.0.zip  
DEMO@dev_demo >
```







## Project Deploy

- Deploys the artifact on a target database (TEST )
  - > `project deploy -file artifact/test-1.0.zip`

```
ADMIN@test_admin > project deploy -file artifact/test-1.0.zip  
Starting the migration...
```





## Commands summary

❑ Initialize a new project

```
project init -name bombshell -schemas HR,CO
```

❑ Export all DB objects

```
project export
```

❑ Export some objects

```
project export -objects HR.COUNTRIES,APEX.101
```

❑ Add custom file/script

```
project stage add-custom -file-name my_file
```

❑ Stage (gen. changelogs/changesets)

```
project stage -branch-name test
```

❑ Release

```
project release -version 1.1
```

❑ Generate artifact (zip)

```
project gen-artifact -name test -version 1.1
```

❑ Deploy

```
project deploy -file artifact/test-1.1.zip
```

# SQLcl Liquibase



Scan to ask a question



## Project (CI/CD) vs. Liquibase

- Liquibase (Open Source) commands are available in SQLcl

```
DEMO@dev_demo > liquibase version
--Starting Liquibase at 2026-03-17T21:00:22.939926 using Java 25.0.2 (version 4.33.0 #0 built at 2025-12-09 17:47+0000)
Liquibase version: 4.33.0 #0 built at 2025-12-09 17:47+0000
Extension Version: 25.4.0.0

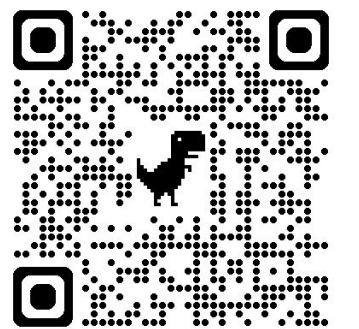
Operation completed successfully.
```

> liquibase | lb [command]

> **help** liquibase

[SQLcl Liquibase documentation](#)

My Liquibase tutorial [here](#) ->



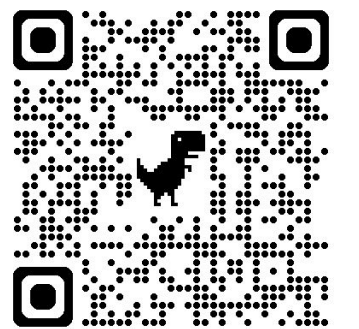




- Use it as you would use standalone Liquibase
- Enhanced to support most of Oracle Database Objects
- Bonus: APEX files support (for free)

[SQLcl Liquibase documentation](#)

My Liquibase tutorial [here](#) ->



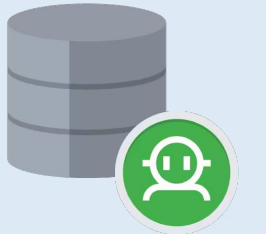


## How Liquibase works?

- Liquibase tracks which database changes has been applied to which environment
- Keeps your database structure consistent across dev, test,prod etc.
- Changes are tracked in DATABASECHANGELOG, and executed SQL can be found in DATABASECHANGELOG\_ACTIONS

```
DEMO@test_demo > select id, filename, dateexecuted,comments, md5sum from databasechangelog;
```





- > sql -name dev\_demo
- > liquibase generate-schema -split -sql

Generates current state of the database into files

```
DEMO@dev_demo > liquibase generate-schema -split -sql
--Starting Liquibase at 2026-03-18T20:13:07.913016 using Java

Export Flags Used:

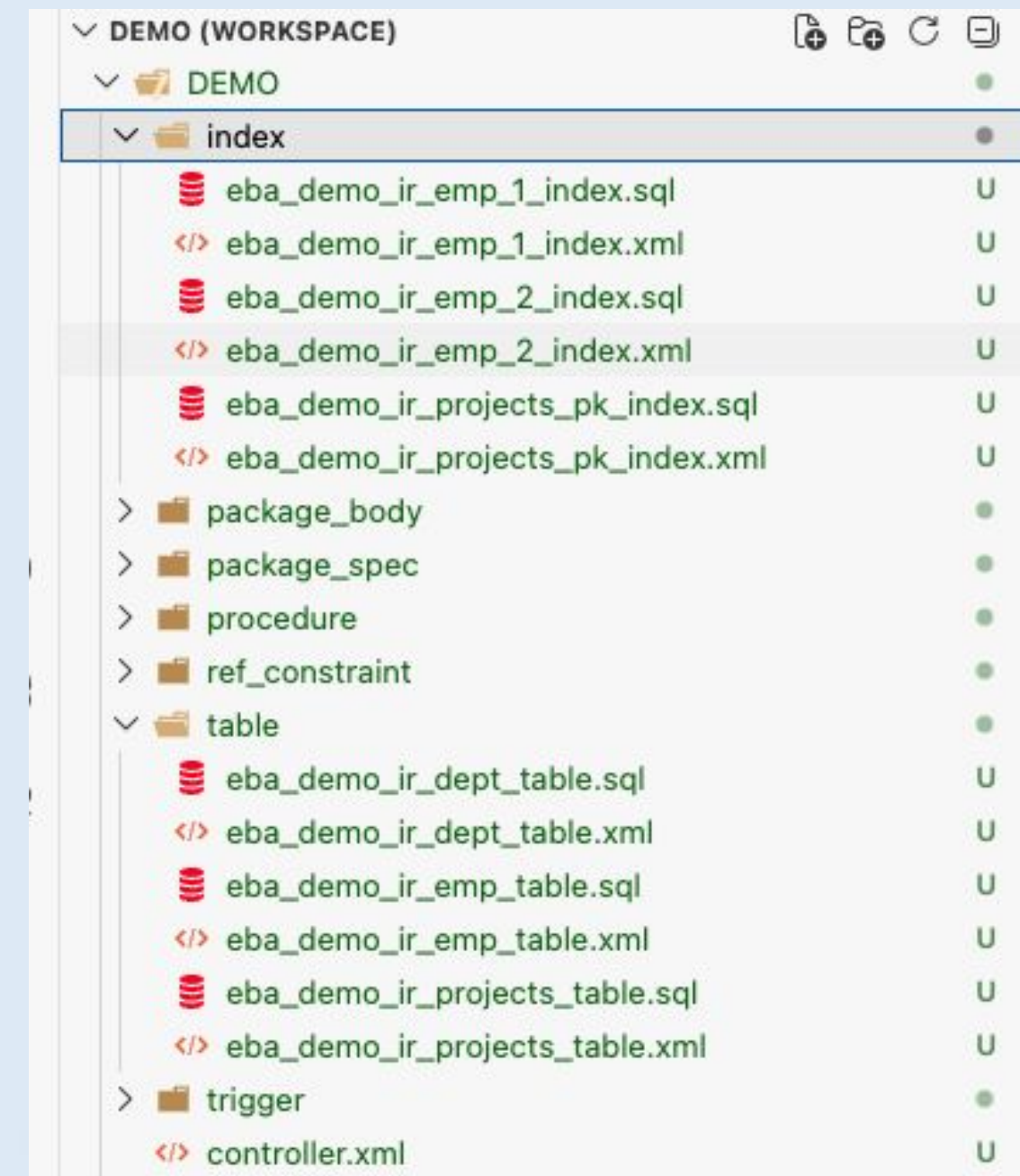
Export Grants      false
Export Synonyms    false

[Method loadCaptureTable]:

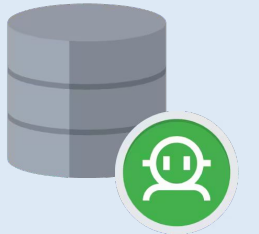
[Method loadCaptureTable]:          1ms
[Load Type DIRECTORY]:             2s 200ms
[Load Type SEQUENCE]:              2s 200ms
[Load Type TYPE_SPEC]:             109ms
[Load Type TYPE_BODY]:             67ms
[Load Type MATERIALIZED_VIEW]:     94ms
[Load Type MATERIALIZED_VIEW_LOG]: 105ms
[Load Type DIMENSION]:             324ms
[Load Type REF_CONSTRAINT]:        407ms
[Load Type CLUSTER]:               15ms
[Load Type FUNCTION]:             390ms
[Load Type PACKAGE_SPEC]:         195ms
[Load Type PROCEDURE]:            39ms
[Load Type VIEW]:                 240ms
[Load Type SYNONYM]:              68ms
[Load Type DB_LINK]:              30ms
[Load Type JOB]:                  335ms
[Load Type PACKAGE_BODY]:         394ms
[Load Type TRIGGER]:              51ms
[Load Type INDEX]:                3s 546ms
[Load Type TABLE]:              10s 904ms
[Method sortCaptureTable]:         283ms
[Method writeChangeLogs]:          59s 506ms
[Method writeController]:         150ms
[Total Time (Wall Clock)]:        1m 1s 377ms
Changelog created and written out to file controller.xml

Operation completed successfully.
```

```
DEMO@dev_demo > |
```







# Apply changes to another environment (TEST)

- > sql -name **test\_demo**
- > liquibase **update** -changelog-file controller.xml

```
</> controller.xml U X
DEMO > </> controller.xml
1  <?xml version="1.0" encoding="UTF-8"?>
2  <databaseChangeLog
3      xmlns="http://www.liquibase.org/xml/ns/dbchangelog"
4      xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5      xmlns:ext="http://www.oracle.com/xml/ns/dbchangelog-ext"
6      xsi:schemaLocation="http://www.liquibase.org/xml/ns/dbchangelog
7          http://www.liquibase.org/xml/ns/dbchangelog/dbchangelog-latest.xsd">
8      <include file="table/eba_demo_ir_dept_table.xml"/>
9      <include file="table/eba_demo_ir_emp_table.xml"/>
10     <include file="table/eba_demo_ir_projects_table.xml"/>
11     <include file="ref_constraint/sys_c0030965_ref_constraint.xml"/>
12     <include file="package_spec/eba_demo_ir_pkg_package_spec.xml"/>
13     <include file="procedure/eba_demo_ir_data_procedure.xml"/>
14     <include file="index/eba_demo_ir_emp_1_index.xml"/>
15     <include file="index/eba_demo_ir_emp_2_index.xml"/>
16     <include file="index/eba_demo_ir_projects_pk_index.xml"/>
17     <include file="trigger/biu_eba_demo_ir_projects_trigger.xml"/>
18     <include file="package_body/eba_demo_ir_pkg_package_body.xml"/>
19 </databaseChangeLog>
```

```
PROBLEMS  TERMINAL
DEMO@test_demo > liquibase update -changelog-file controller.xml
--Starting Liquibase at 2026-03-18T20:25:48.759223 using Java 25.0.2 (version 4.33.0 #0 built at 2025-12-09 17:47+0000)
Running Changeset: table/eba_demo_ir_dept_table.xml::5a1d44dbc39d1b71208a40d44532e604c12a62f2::(DEMO)-Generated
Table "EBA_DEMO_IR_DEPT" created.
Running Changeset: table/eba_demo_ir_emp_table.xml::4b41dd083ed44f7d1b26bb2ea0353af6fabe9843::(DEMO)-Generated
[]
```



## Generates current state of the chosen object

> liquibase **generate-db-object** -obn eba\_demo\_ir\_dept -obt table -sql



```
PROBLEMS  TERMINAL

DEMO@dev_demo > liquibase generate-db-object -object-name eba_demo_ir_dept -object-type table -sql
```

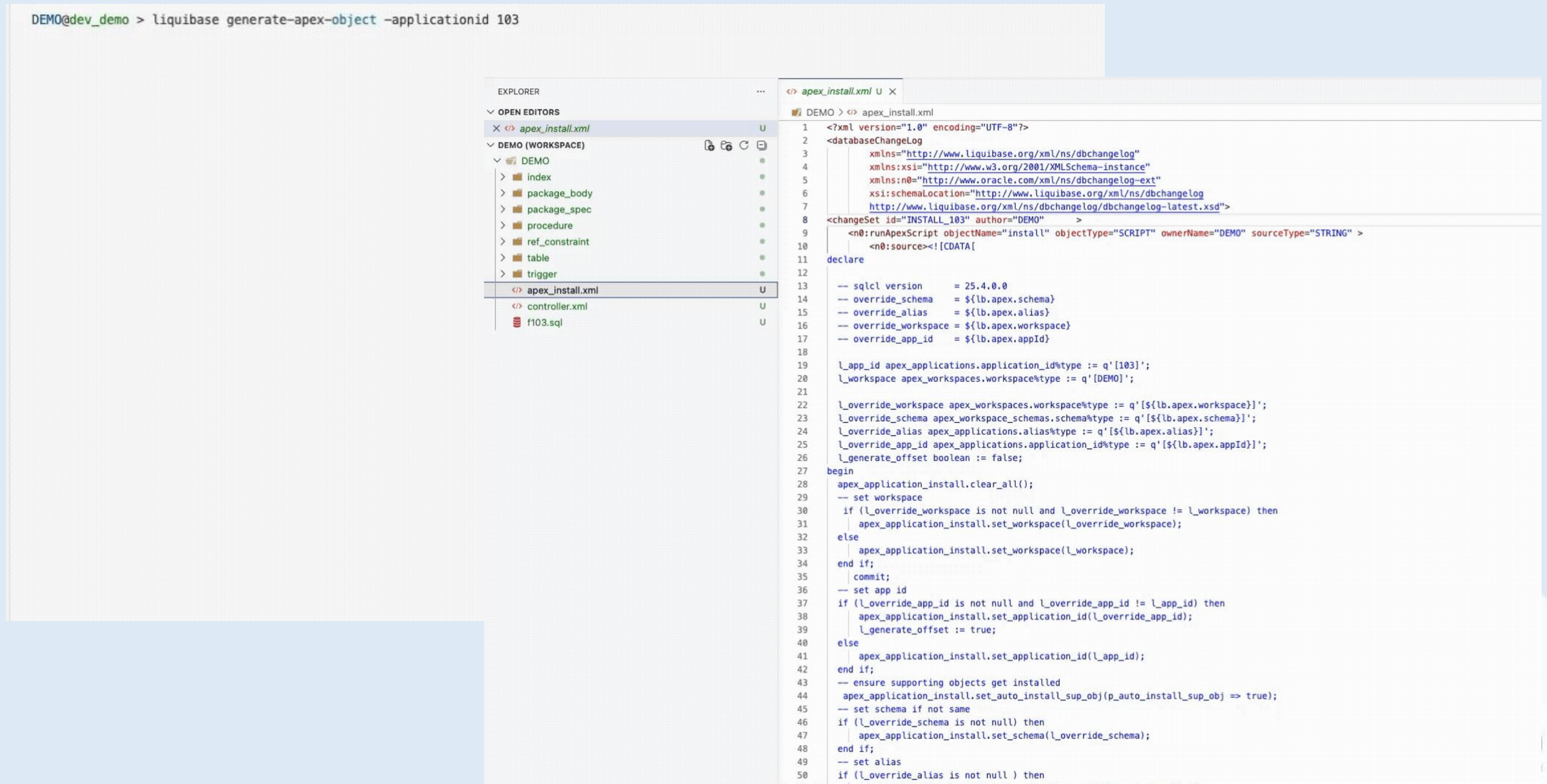




# Generate APEX changeset

> liquibase generate-apex-object -applicationid 103

```
DEMO@dev_demo > liquibase generate-apex-object -applicationid 103
```



```
<?xml version="1.0" encoding="UTF-8"?>
<databaseChangeLog
  xmlns="http://www.liquibase.org/xml/ns/dbchangelog"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ext="http://www.oracle.com/xml/ns/dbchangelog-ext"
  xsi:schemaLocation="http://www.liquibase.org/xml/ns/dbchangelog
    http://www.liquibase.org/xml/ns/dbchangelog/dbchangelog-latest.xsd">
  <changeSet id="INSTALL_103" author="DEMO" >
    <n0:runApexScript objectName="install" objectType="SCRIPT" ownerName="DEMO" sourceType="STRING" >
      <n0:source><![CDATA[
declare
  -- sqlcl version      = 25.4.0.0
  -- override_schema    = ${lb.apex.schema}
  -- override_alias     = ${lb.apex.alias}
  -- override_workspace = ${lb.apex.workspace}
  -- override_app_id    = ${lb.apex.appId}

  l_app_id apex_applications.application_id%type := q'[103]';
  l_workspace apex_workspaces.workspace%type := q'[DEMO]';

  l_override_workspace apex_workspaces.workspace%type := q'[${lb.apex.workspace}]';
  l_override_schema apex_workspaces_schemas.schema%type := q'[${lb.apex.schema}]';
  l_override_alias apex_applications.alias%type := q'[${lb.apex.alias}]';
  l_override_app_id apex_applications.application_id%type := q'[${lb.apex.appId}]';
  l_generate_offset boolean := false;

begin
  apex_application_install.clear_all();
  -- set workspace
  if (l_override_workspace is not null and l_override_workspace != l_workspace) then
    apex_application_install.set_workspace(l_override_workspace);
  else
    apex_application_install.set_workspace(l_workspace);
  end if;
  commit;
  -- set app id
  if (l_override_app_id is not null and l_override_app_id != l_app_id) then
    apex_application_install.set_application_id(l_override_app_id);
    l_generate_offset := true;
  else
    apex_application_install.set_application_id(l_app_id);
  end if;
  -- ensure supporting objects get installed
  apex_application_install.set_auto_install_sup_obj(p_auto_install_sup_obj => true);
  -- set schema if not same
  if (l_override_schema is not null) then
    apex_application_install.set_schema(l_override_schema);
  end if;
  -- set alias
  if (l_override_alias is not null ) then
    apex_application_install.set_application_alias(l_override_alias);
  end if;
end;
]]]>
    </n0:runApexScript>
  </changeSet>
</databaseChangeLog>
```





## SQLcl Liquibase is super smart

- It exactly knows if change should be deployed or not
- Example:
  - I added a new column NEW\_COL1 into **EBA\_DEMO\_IR\_DEPT** table at DEV
  - Meanwhile someone added same column directly at TEST (ignoring CI/CD process)



# SQLcl Liquibase is super smart

```
DEMO@test_demo > lb update -chf eba_demo_ir_dept_table.xml
--Starting Liquibase at 2026-03-21T13:09:56.054994 using Java 25.0.2 (version 4.33.0 #0 built at 2025-12-09 17:47+0000)
Running Changeset: eba_demo_ir_dept_table.xml::78acf6fd6d2b80889b133b4da651db4e706c5c58::(DEMO)-Generated

UPDATE SUMMARY
Run: 1
Previously run: 0
Filtered out: 0
-----
Total change sets: 1

Liquibase: Update has been successful. Rows affected: 0

Operation completed successfully.

DEMO@test_demo > select filename, sql, status from databasechangelog_actions order by deployment_id desc fetch first 1 row only;
```

| FILENAME                   | SQL                                        | STATUS |
|----------------------------|--------------------------------------------|--------|
| eba_demo_ir_dept_table.xml | -- Objects are the the same, no work to do | RAN    |

```
DEMO@test_demo > |
```

# DIFF - compare two schemas and synchronize them



Scan to ask a question



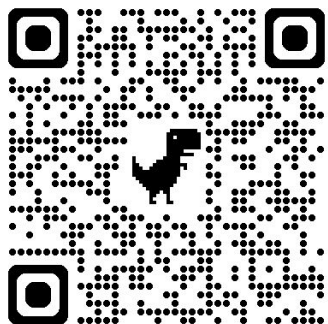


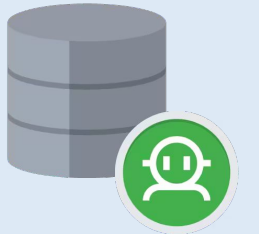
# DIFF

- Use it to compare two Oracle DB schemas and generate SQL scripts (artifact) with necessary changes to synchronize them
- Extremely helpful if you struggle with database schema drift (environments are not equal)
- Use DIFF before you start using CI/CD Tools ( SQLcl Projects, Liquibase etc.)

[DIFF documentation](#)

More about DIFF at Tomas Kucharzyk's blog [here](#) ->





SOURCE

DEV [dev\_demo]

- 12 database objects

TARGET

TEST [test\_demo]

- 11 database objects

> **diff -source dev\_demo -target test\_demo**

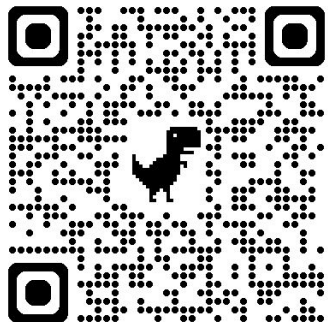
```
@ > diff -source dev_demo -target test_demo
[diff - info] Preparing clean project directory: /var/folders/dj/nwvnq0jn5992_8rxcbcv5xbr0000gn/T/diff_project_560707579396177596
[diff - info] Initializing project with schemas: DEMO,TEST
[diff - info] Generating diffs via project stage against: base_DEMO
[diff - info] Generating artifact (SQLcl default location)
[diff - info] Artifact copied from "/var/folders/dj/nwvnq0jn5992_8rxcbcv5xbr0000gn/T/diff_project_560707579396177596/artifact/diff DEMO vs TEST.zip" to "/Users/rg/git/sandbox/demo_sqlcl/diff/diff_DEMO_vs_TEST-1779711258209.zip"
[diff - info] Success. Diffs under dist/releases/next. Logs at: null
@ > |
```



# Review generated SQL scripts

- Unzip and go to releases/next/changes/target\_test and review generated files

More about DIFF at Tomas Kucharzyk's blog [here](#) ->







EXPLORER

...

> OPEN EDITORS

▼ DEMO\_SQLCL

> apex

▼ diff

▼ diff\_DEMO\_vs\_TEST-1779711258209

> env

▼ releases

▼ next

▼ changes

▼ target\_test

▼ demo

> comments

> functions

> package\_specs

> procedures

> tables

> triggers

stage.changelog.sql

release.changelog.sql

main.changelog.sql

> utils

install.sql

stage.changelog.sql X

diff > diff\_DEMO\_vs\_TEST-1779711258209 > releases > next > changes > targ

1 --stage

2 @@demo/triggers/loki\_before\_ddl\_tgr.sql

3 @@demo/procedures/eba\_demo\_ir\_data.sql

4 @@demo/functions/perf\_sample\_get\_employee\_status.sql

5 @@demo/package\_specs/sqlcl\_lb\_capture.sql

6 @@demo/package\_specs/eba\_demo\_ir\_pkg.sql

7 @@demo/comments/esql\_1220.sql

8 @@demo/tables/esql\_1220.sql

9 @@demo/tables/eba\_demo\_ir\_projects.sql

10 @@demo/tables/eba\_demo\_ir\_emp.sql

11 @@demo/tables/eba\_demo\_ir\_dept.sql

12 @@demo/tables/dbtools\$mcp\_log.sql

13

PROBLEMS

OUTPUT

TERMINAL

CHAT

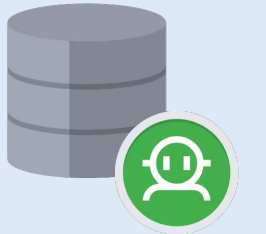
@ >

OATUG

OHUG

ASCEND  
2026

<<time>>



```
> sql -name test_demo
```

```
> project deploy -file [your_filename].zip
```

The screenshot shows the Visual Studio Code interface. On the left, the Explorer sidebar displays the project structure under 'DEMO\_SQLCL'. It includes folders 'apex', 'diff', and 'project' (which is selected). The 'diff' folder contains 'diff\_DEMO\_vs\_TEST-1779711258209' (a folder) and 'demo\_diff.md' (a file). The 'project' folder is currently empty. Below the Explorer is the 'SCRIPTS' section, showing 'create\_user.sql' (a file) and 'show\_level\_and\_scripts' (a folder) containing 'README.md' (a file). On the right, the main editor area is mostly blank with a large, faint 'X' watermark. At the bottom, the 'TERMINAL' tab is active, showing the command: `TEST@test_demo > project deploy -file diff_DEMO_vs_TEST-1779711258209.zip`. The terminal output area above the command is empty.

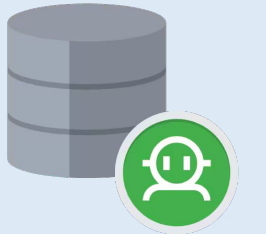
pretius

# SQLcl Codescan



Scan to ask a question





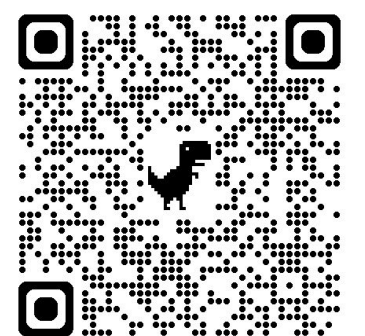
# What is CODESCAN?

- Checks quality of written PL/SQL and SQL
- Follows the rules from Trivadis PL/SQL & SQL Coding Guidelines
- Identifies performance issues stated in Oracle documentation

[SQL Performance Issues](#)

[Trivadis PL/SQL & SQL Coding Guidelines](#)

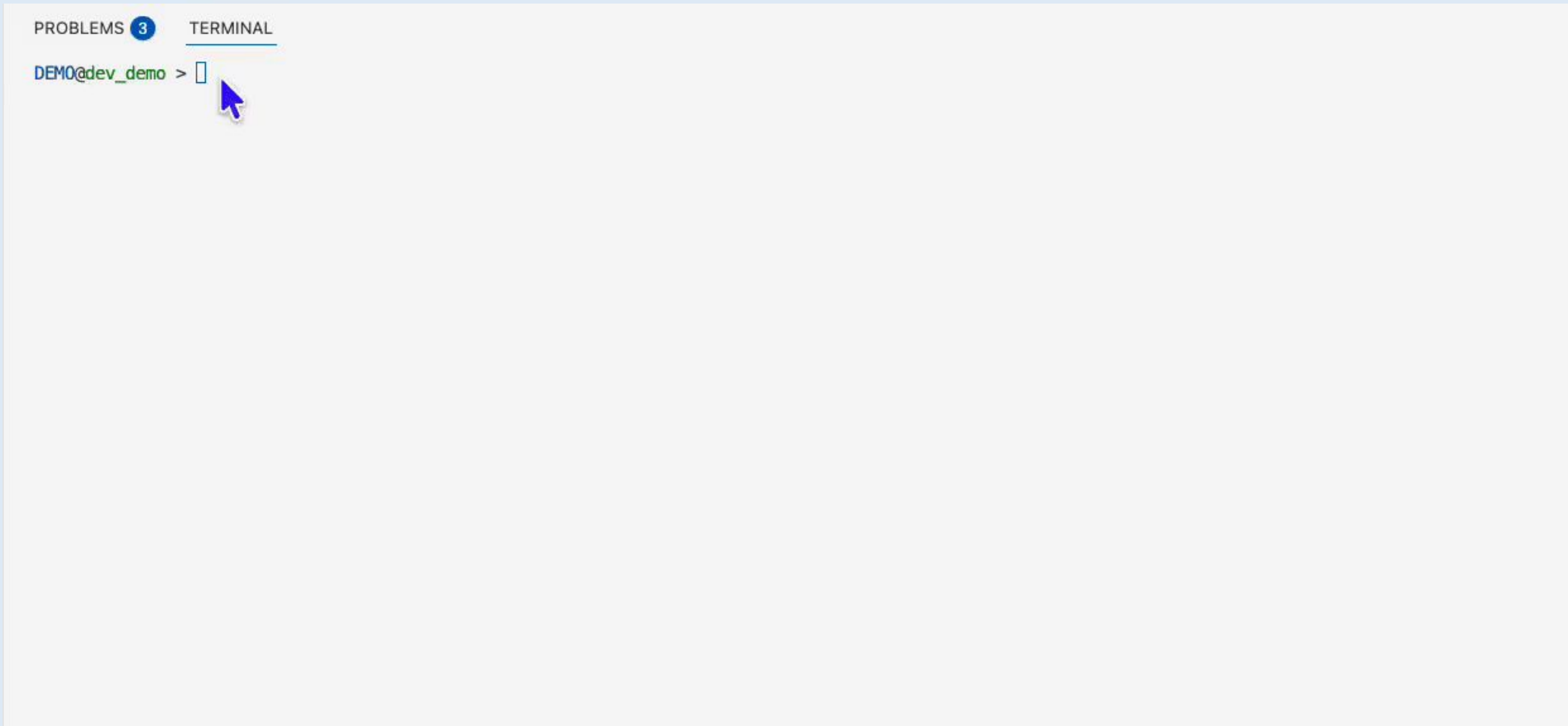
My blog about CODESCAN [here](#) ->





# LIVE quality scan

> set **codescan** on [default off]



# LIVE scan of executed scripts



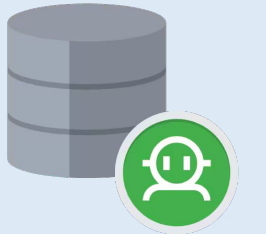
```
my_script.sql 9, U X
DEMO > my_script.sql > ...
1  --G-1030 "Avoid defining variables that are not used"
2  DECLARE
3    l_dummy NUMBER := 1;
4  BEGIN
5    NULL;
6  END;
7  /
8  --G-3145 "Avoid using SELECT * directly from a table or view"
9  SELECT *
10 FROM eba_demo_ir_emp
11 WHERE ename = 'KING';
12
13 --SQL best practice warning (1,19): G-7410: Avoid standalone functions put your functions in packages
14 --SQL best practice warning (1,60): G-7160: Always explicitly state parameter mode
15 CREATE OR REPLACE FUNCTION perf_sample_get_employee_status(p_empno VARCHAR2)
16 RETURN VARCHAR2 IS
17 BEGIN
18   RETURN 'STATUS_' || p_empno;
19 END perf_sample_get_employee_status;
20 /

TERMINAL

DEMO@dev_demo > |
```

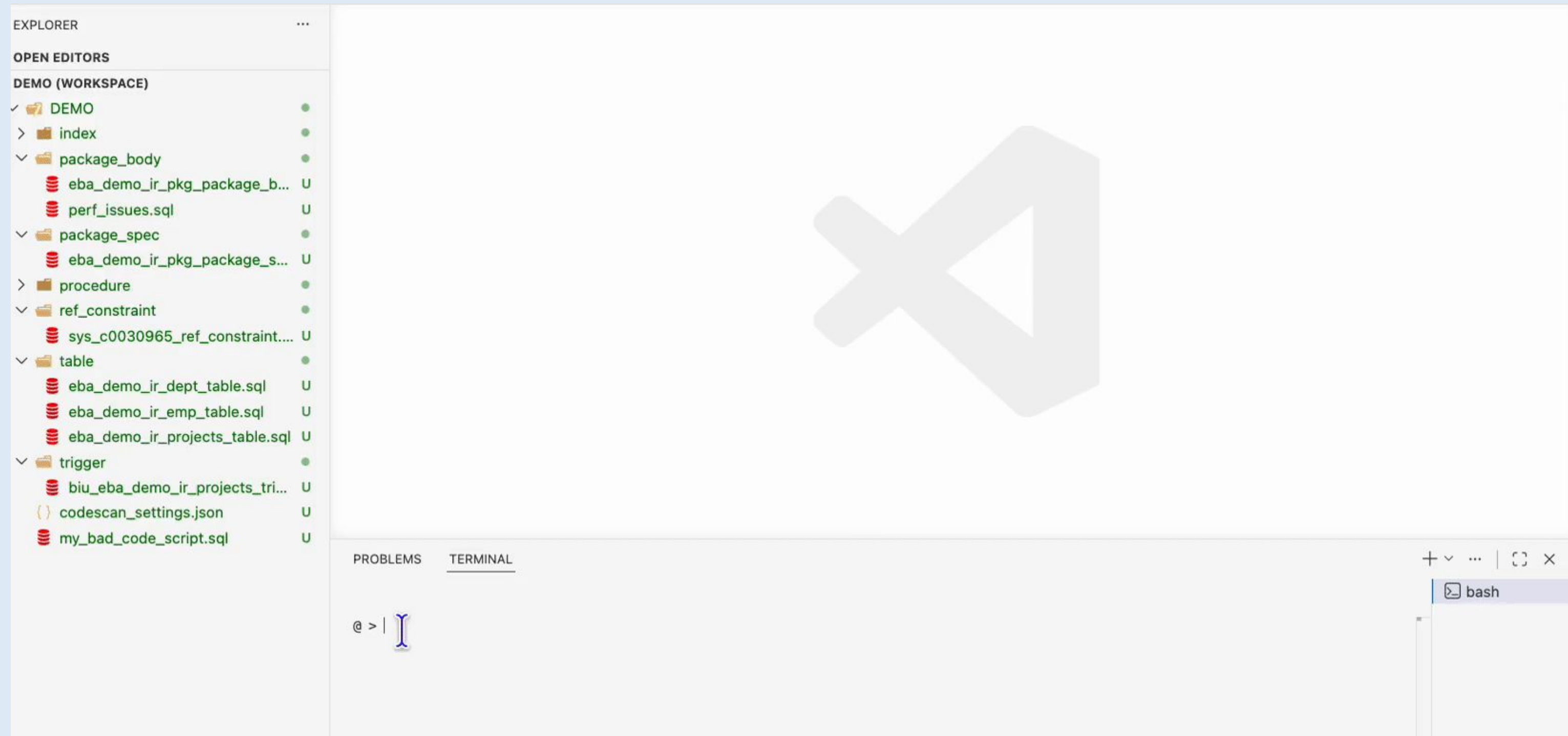
00:00:00.660

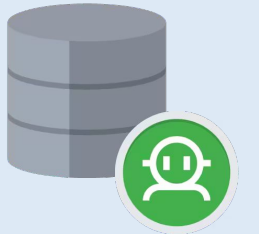




# Scanning multiple files (**connection not required!**)

> **codescan** -format json -output my\_default\_scan.json





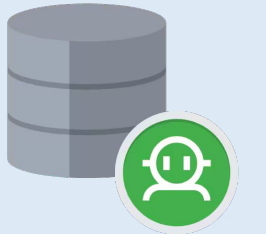
# Ignore rules you don't like / don't agree with

```
{ } my_default_scan.json U X
DEMO > { } my_default_scan.json > { } 0 > { } issues
1  [
2  {"file": "/Users/rg/DEMO/my_bad_code_script.sql", "issues": [
3    {"line": 2, "col": 2, "ruleNo": "G-1030", "msg": "Avoid defining variables that are not used"},
4    {"line": 8, "col": 7, "ruleNo": "G-3145", "msg": "Avoid using SELECT * directly from a table or view"},
5    {"line": 14, "col": 18, "ruleNo": "G-7410", "msg": "Avoid standalone functions put your functions in packages"},
6    {"line": 14, "col": 59, "ruleNo": "G-7160", "msg": "Always explicitly state parameter mode"},
7    {"line": 24, "col": 1, "ruleNo": "G-3130", "msg": "Try to use ANSI SQL-92 join syntax"},
8    {"line": 25, "col": 18, "ruleNo": "PSR-203", "msg": "An inline view or table detected that was OUTER-joined on optional side of the joi"},
9    {"line": 34, "col": 1, "ruleNo": "G-5060", "msg": "Avoid unhandled exceptions"},
10   {"line": 36, "col": 9, "ruleNo": "PSR-102", "msg": "Scalar subqueries are subqueries in SELECT list. They must return exactly one value"},
11   {"line": 42, "col": 1, "ruleNo": "G-5010", "msg": "Try to use a error/logging framework for your application"},
12   {"line": 42, "col": 1, "ruleNo": "G-7510", "msg": "Always prefix Oracle supplied packages with owner schema name"}
13 ]},
14 {"file": "/Users/rg/DEMO/trigger/biu_eba_demo_ir_projects_trigger.sql", "issues": [
15   {"line": 2, "col": 7, "ruleNo": "G-7730", "msg": "Avoid multiple DML events per trigger"},
16   {"line": 6, "col": 12, "ruleNo": "G-9030", "msg": "Try to define a default value on conversion errors"},
17   {"line": 6, "col": 69, "ruleNo": "G-5060", "msg": "Avoid unhandled exceptions"},
18   {"line": 14, "col": 32, "ruleNo": "G-5050", "msg": "Avoid use of the RAISE_APPLICATION_ERROR built-in procedure with a hard-coded 20nnn"}
19 ]},
20 {"file": "/Users/rg/DEMO/package_body/eba_demo_ir_pkg_package_body.sql", "issues": [
21   {"line": 14, "col": 34, "ruleNo": "G-2160", "msg": "Avoid initializing variables using functions in the declaration section"},
22   {"line": 21, "col": 41, "ruleNo": "G-1050", "msg": "Avoid using literals in your code"},
23   {"line": 22, "col": 41, "ruleNo": "G-1050", "msg": "Avoid using literals in your code"},
24 ]}
```

```
{ } ignored_rules.json U X
DEMO > { } ignored_rules.json > ...
1  {
2    "ignore": {
3      "G-3145",
4      "G-3130",
5      "G-1050"
6    }
7  }
```

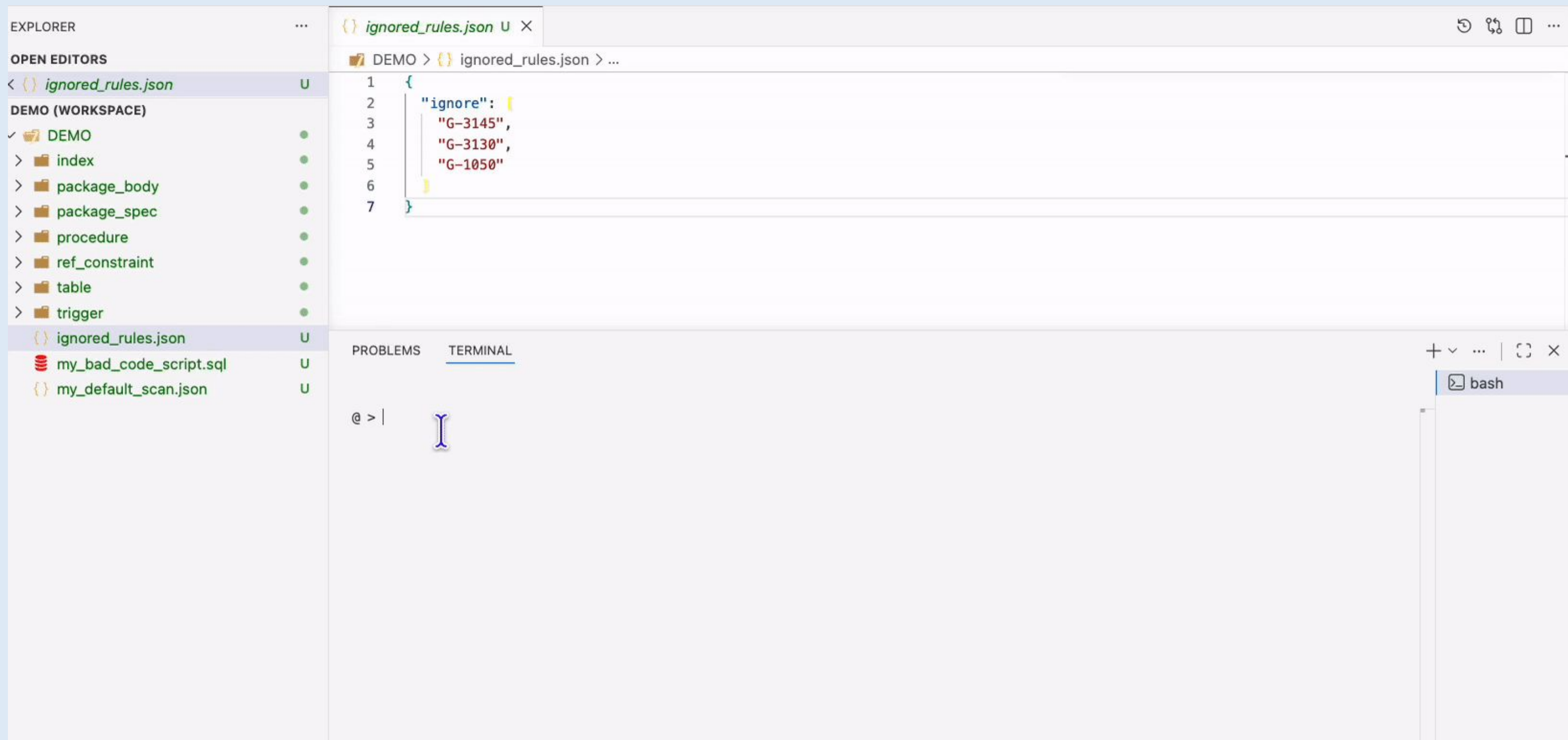
\* Ignore doesn't work with Performance Rules (PSR-XXX)





# Ignore rules you don't like / don't agree with

> **codescan** -format json -output custom\_scan.json -**settings** ignored\_rules.json





pretius

# SQLcI MCP + AI



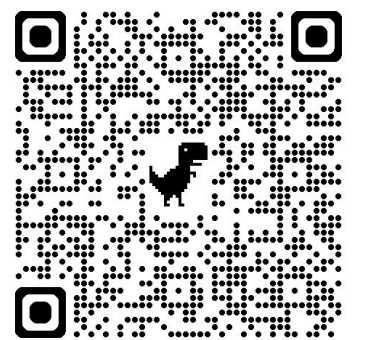
Scan to ask a question



# What is MCP?

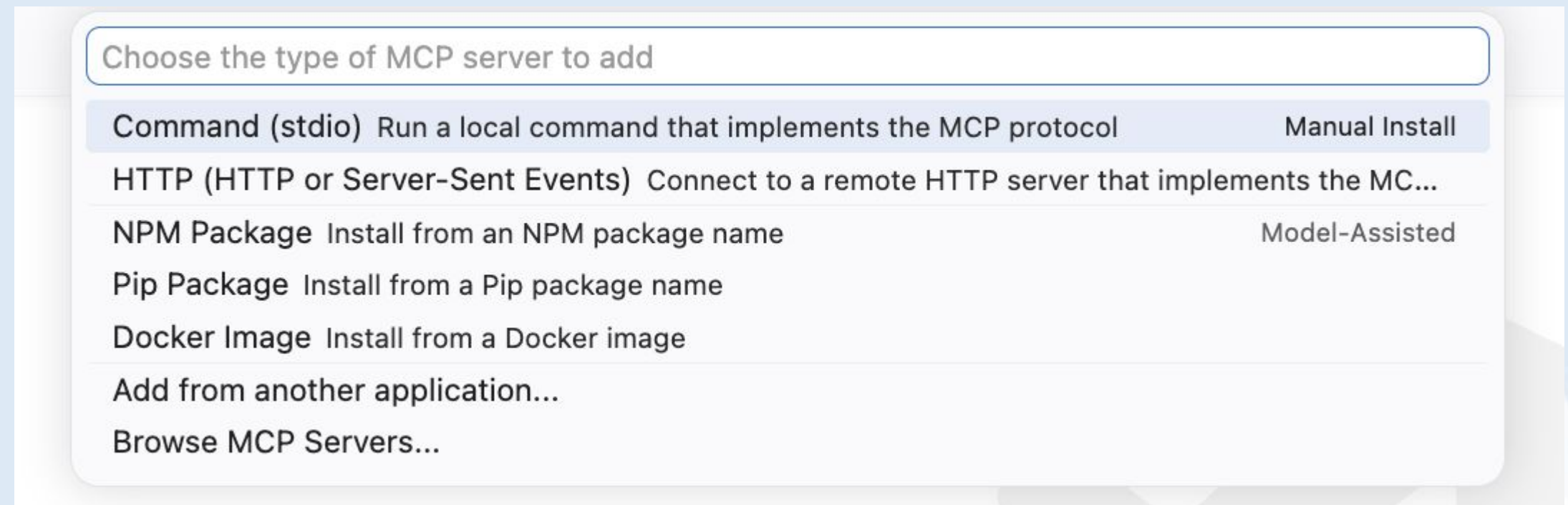
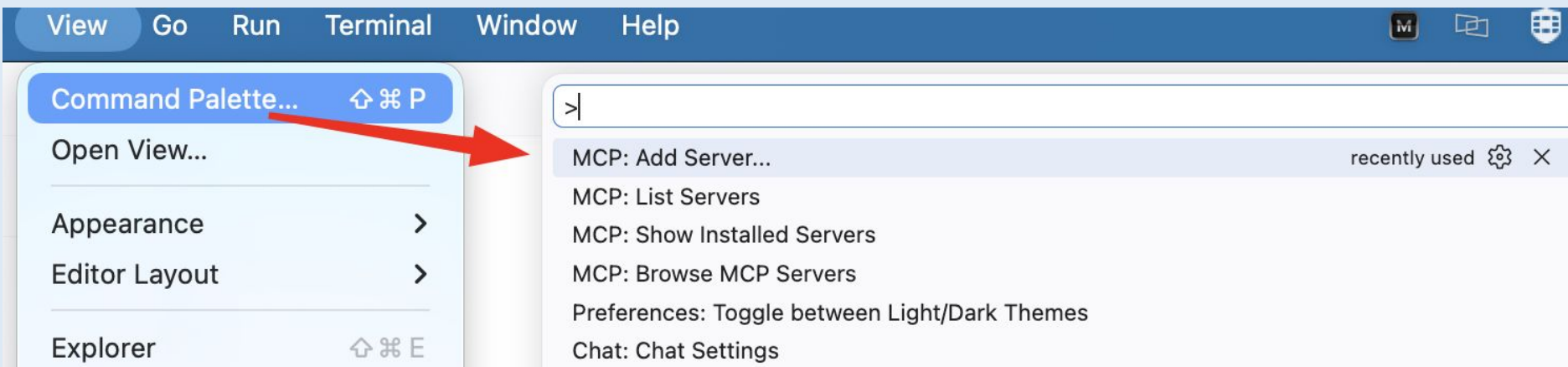
- **Model Context Protocol** is a standard that allows AI models to connect with external data e.g databases

My intro blog about MCP [here](#) ->





## Add MCP in VS Code







## Add MCP in VS Code

Enter Command

/Users/rg/apps/oracle/sqlcl/bin/sql|

Press 'Enter' to confirm your input or 'Escape' to cancel

Enter Server ID

conference\_mcp|

Press 'Enter' to confirm your input or 'Escape' to cancel



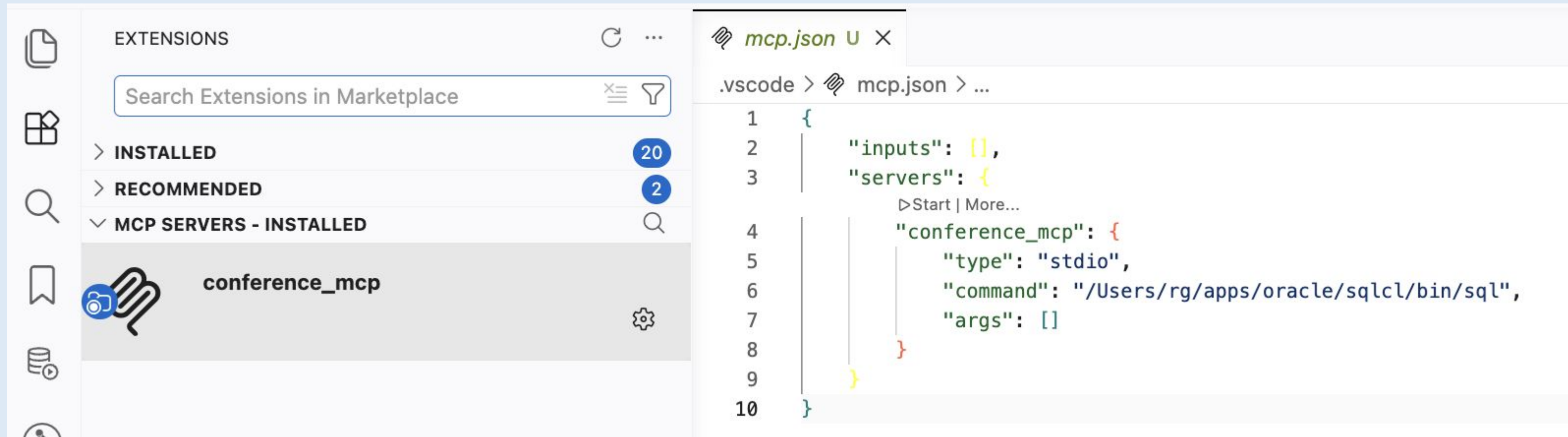
## Add MCP in VS Code

### Add MCP Server

Select the configuration target

Global Available in all workspaces, runs locally

Workspace Available in this workspace, runs locally



Configure Tools

Select tools that are available to chat.47 SelectedOK

The selected tools will be applied globally for all chat sessions that use the default agent.

Built-In

search Search files in your workspace

todo Manage and track todo items for task planning

vscode Use VS Code features

web Fetch information from the web

conference\_mcp

connect Provides an interface to connect to a specified database. If a database connecti...

dbtools-get-tool-request Retrieves the current status and (if available) the final result o...

disconnect This tool performs a disconnection from the current session in an Oracle dat...

list-connections Retrieves a list of connection names. A connection name can be used t...

run-sql This tool executes SQL queries in an Oracle database. If no active connection exi...

run-sqlcl This tool executes SQLcl commands in the SQLcl CLI. If the given command req...

schema-information This tool provides detailed insights about a specified schema by an...

Update Tools

GitKraken

SQLcl - SQL Developer

Describe what to build

Agent

Auto

# SQLcl + MCP AI

<<time>>

OATUGOHUGASCEND2026





EXTENSIONS

Search Extensions in Marketplace

> INSTALLED 20

> ENABLED 19

✓ MCP SERVERS - INSTALLED

conference\_mcp

Start Server

Stop Server

Restart Server

Enable

Enable (Workspace)

Disable

Disable (Workspace)

Restart Server

Show Output

Show Configuration

Show Configuration (JSON)

Configure Model Access

Show Sampling Requests

Browse Resources

Uninstall

Install in Workspace



SQL DEVELOPER

> SQL SNIPPETS

> CONNECTIONS

> customers

> priv

> demo

> dev\_admin

> dev\_demo

> test\_admin

> test\_demo

PROBLEMS

OUTPUT

TERMINAL

CHAT

+ v ↺ ⚙ ... | [ ] ×

CONNECTING TO DEV\_DEMO DATABASE ON CONFERENCE\_MCP SERVER

Connect to my dev\_demo database using conference\_mcp server.

You want an active DB session to `dev_demo` via the conference MCP integration, so I'll attempt a direct connection first and then report the status or next required step if the named connection is missing.

Run `connect` – SQLcl - SQL Developer (MCP Server)

Provides an interface to connect to a specified database. If a database connection is already active, prompt the user for confirmation before switching to the new connection. If no connection exists, list the available schemas for selection.  
the connection name is case sensitive  
Note: If the provided connection is invalid or does not match any saved connection, display instructions to the user on how to create a named connection in SQLcl  
The `model` argument should specify only the actual name and version of the LLM (Large Language Model) you are using, with no additional information

Input

{ "connection\_name": "dev\_demo", "model": "GPT-5.3-Codex" }

See more

Note that MCP servers or malicious conversation content may attempt to misuse 'Code' through tools.

Allow in this Session | ⌵ Skip



CONNECTING TO DEV\_DEMO VIA CONFERENCE\_MCP SERVER



Connect to "dev\_demo" using my mcp server called "conference\_mcp"

I'll attempt the connection now using the `dev_demo` saved profile on your active SQLcl MCP tool context, then confirm whether it's connected.

Ran `connect` – SQLcl – SQL Developer (MCP Server)

Connected to `dev_demo` successfully via your active MCP SQLcl server context ( `conference_mcp` ).

Connection status:

- `dev_demo` : connected
- Oracle DB version: `23.26.1.2.0`



GPT-5.3-Codex • 0.9x





CONNECTING TO DEV\_DEMO VIA CONFERENCE\_MCP SERVER

Restore Checkpoint ·

Give me a brief overview of my database structure

Optimizing tool selection...

Considered database structure overview

Run `schema-information` – SQLcl - SQL Developer (MCP Server)

This tool provides detailed insights about a specified schema by analyzing and enriching the metadata returned from the executed queries.

It supports synchronous and asynchronous execution types via the `executionType` argument.

The tool accepts the following arguments:

[Show More](#)

Input

```
{ "level": "BRIEF", "model": "Claude Haiku 4.5" }
```

See more

*Note that MCP servers or malicious conversation content may attempt to misuse 'Code' through tools.*

Allow in this Session | Skip



Organized schema information into a brief overview

Here's a brief overview of your **DEMO** schema:

#### Core Tables:

- **EBA\_DEMO\_IR\_EMP** — Employee data (EMPNO, ENAME, JOB, MGR, HIREDATE, SAL, COMM, DEPTNO)
- **EBA\_DEMO\_IR\_DEPT** — Department data (DEPTNO, DNAME, LOC, CONFERENCE\_ID)
- **EBA\_DEMO\_IR\_PROJECTS** — Project tracking (PROJECT, TASK\_NAME, START\_DATE, END\_DATE, STATUS, ASSIGNED\_TO, COST, BUDGET)
- **DBTOOLS\$MCP\_LOG** — MCP logging table
- **ESQL\_1220** — Migration/utility table with extensive metadata columns

**Indexes:** 7 indexes (mostly supporting primary keys and foreign key relationships)

#### Code Objects:

- 2 Packages: **EBA\_DEMO\_IR\_PKG**, **SQLCL\_LB\_CAPTURE** (with bodies)
- 1 Procedure: **EBA\_DEMO\_IR\_DATA**
- 1 Function: **PERF\_SAMPLE\_GET\_EMPLOYEE\_STATUS**
- 1 Trigger: **BIU\_EBA\_DEMO\_IR\_PROJECTS**

**Other:** Sequences, LOBs for handling large objects

This looks like a demonstration/conference database with employee/department/project management tables plus MCP tooling infrastructure.





## MCP and security?

- Assing minimum permissions
  - create db user account with minimum permissions required for its tasks
- Avoid production database access
  - use sanitized data, read-only copies or dedicated data subsets
- Audit LLM activity
  - auto-created DBTOOLS\$MCP\_LOG table for activity logging





# Configure restrict levels for SQLcl MCP Server

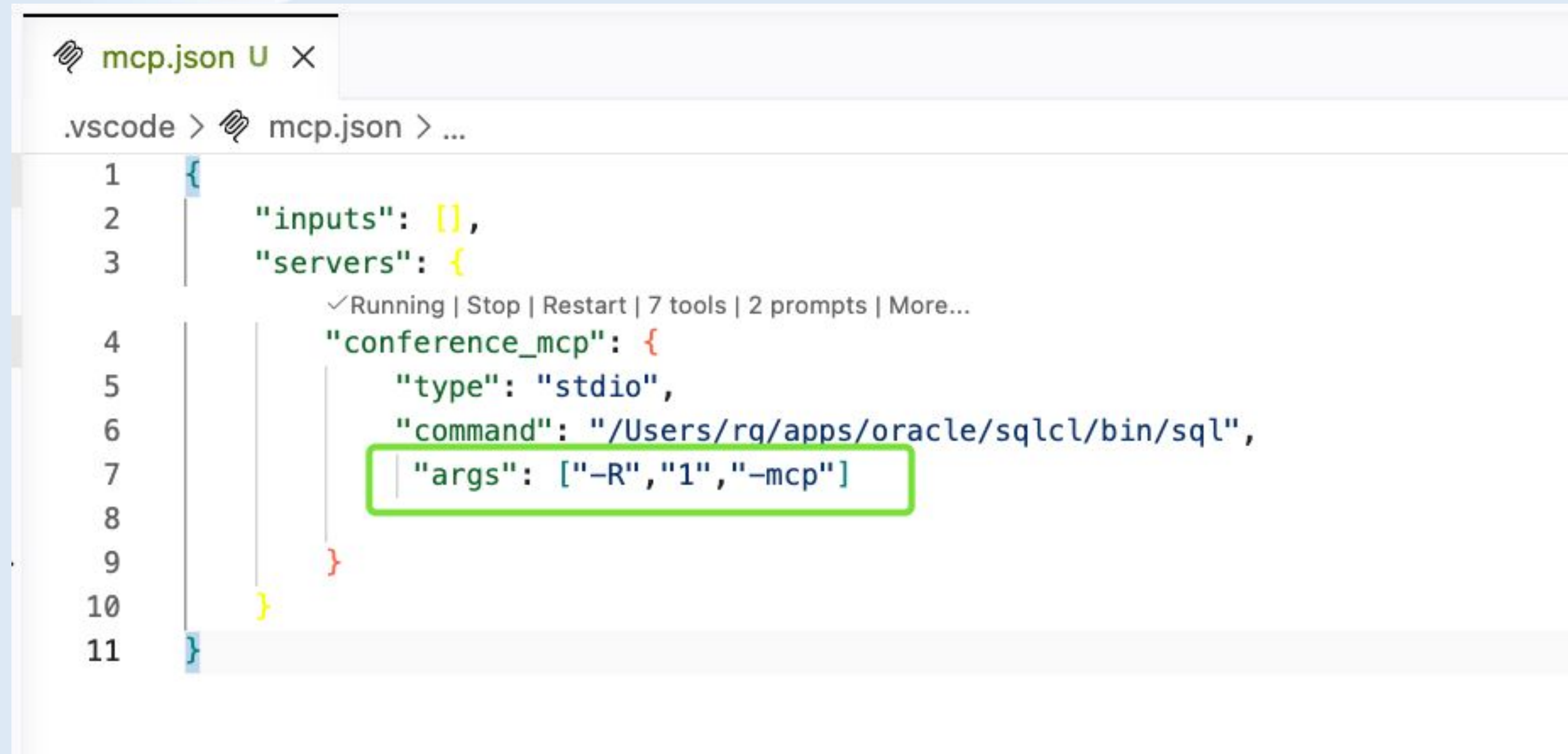
| Level       | Access                |                                                                   |
|-------------|-----------------------|-------------------------------------------------------------------|
| 4 (default) | Most restrictive      | Blocks all command from level 1-4                                 |
| 3           | Highly restrictive    | Disable running external scripts in SQLcl ( @, @@)                |
| 2           | Medium restrictive    | Prevent saving SQL results to external files (SAVE, SPOOL, STORE) |
| 1           | Minimally restrictive | Prevents HOST or EDIT commands                                    |
| 0           | Unrestricted          | Allows all commands                                               |

[Commands Restricted by Level](#)

[Configuring Restrict Levels](#)



## Configure restrict levels for SQLcl MCP Server



The screenshot shows a VS Code editor window with a file named `mcp.json`. The editor displays the following JSON configuration:

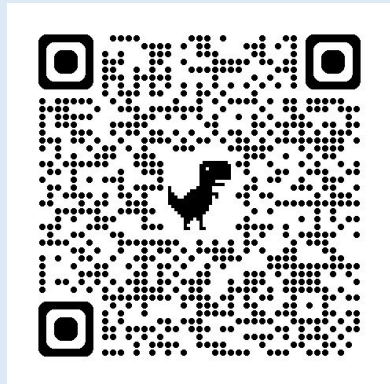
```
1 {
2   "inputs": [],
3   "servers": {
4     "conference_mcp": {
5       "type": "stdio",
6       "command": "/Users/rq/apps/oracle/sqlcl/bin/sql",
7       "args": ["-R", "1", "-mcp"]
8     }
9   }
10 }
11 }
```

A green rectangular box highlights the `"args": ["-R", "1", "-mcp"]` line in the configuration. Above the `"conference_mcp"` block, a status bar indicates: `✓ Running | Stop | Restart | 7 tools | 2 prompts | More...`.

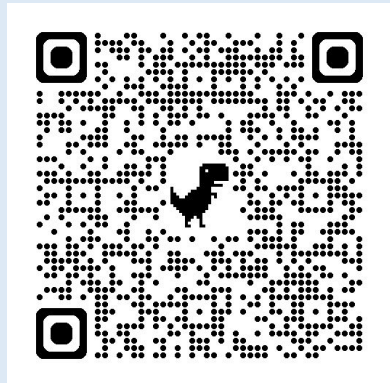


Worth reading:

[Getting Started with our MCP Server for Oracle Database \(by Jeff Smith\)](#)



[SQLcl MCP Server Official Documentation](#)





# Install AI DBTools Skills via SQLcl



Scan to ask a question



## List existing AI skills

```
> sql /nolog  
> skills list
```

```
@ > skills list -v  
INFO: Listing DBTools skills from /Users/rg/.dbtools/skills/repositories/oracle-skills.  
WARN: No local DBTools skills repository found. Run skills sync to synchronize it.  
@ > |
```





# Synchronize skills

> **skills sync**

```
@ > skills sync
SUCCESS: Synchronized DBTools skills repository to /Users/rg/.dbtools/skills/repositories/oracle-skills version latest (refs/he
ads/main -> 11e528de5ea0686b53adf1d853c909a4bf9ed4d9).
SUCCESS: Installed 15 entries into agent skill locations.
WARN: Skipped existing skill entries: /Users/rg/.copilot/skills/apex, /Users/rg/.copilot/skills/db, /Users/rg/.copilot/skills/f
usion, /Users/rg/.copilot/skills/graal, /Users/rg/.copilot/skills/oci
SUCCESS: DBTools skills sync log: /Users/rg/.dbtools/skills/logs/skills-sync-20260525T104313.345Z.log
@ >
```

```
@ > skills list -v
INFO: Listing DBTools skills from /Users/rg/.dbtools/skills/repositories/oracle-skills.
+=====+
+ Skill  + Status   + Agents                                     +
+=====+
| apex   | INSTALLED | cline, dexto, github-copilot, warp |
+-----+
| db     | INSTALLED | cline, dexto, github-copilot, warp |
+-----+
| fusion | INSTALLED | cline, dexto, github-copilot, warp |
+-----+
| graal  | INSTALLED | cline, dexto, github-copilot, warp |
+-----+
| oci    | INSTALLED | cline, dexto, github-copilot, warp |
+-----+
```





# AI skills for SQLcl

SQLCL USAGE SKILLS

Start Over

Do you have configured skills on how to use SQLcl ?

Read skill db

Yes. The db skill set includes a SQLcl section under `db/sqlcl/` with guides on basics, scripting, Liquibase integration, formatting, and MCP server usage. If you tell me what you need (e.g., saved connections, scripting patterns, or MCP setup), I will pull the specific SQLcl skill file and answer precisely.

GPT-5.2-Codex • 1x

| Name                                                                                  |                           |
|---------------------------------------------------------------------------------------|---------------------------|
|    | sqlcl-awr.md              |
|    | sqlcl-background-jobs.md  |
|    | sqlcl-basics.md           |
|    | sqlcl-cicd.md             |
|   | sqlcl-data-loading.md     |
|  | sqlcl-ddl-generation.md   |
|  | sqlcl-formatting.md       |
|  | sqlcl-liquibase.md        |
|  | sqlcl-mcp-server.md       |
|  | sqlcl-scheduler-daemon.md |
|  | sqlcl-scripting.md        |

# Customise SQLcl



Scan to ask a question



## Display username & connection name

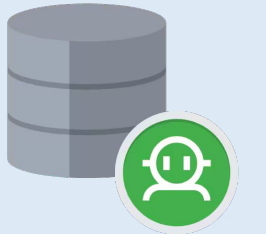
```
DEMO@dev_demo > select 1 from dual;
```

```
1
```

```
-----  
1
```

```
> set sqlprompt "@|blue _USER|@@@|green _CONNECT_IDENTIFIER|@ > "
```





## Display status bar

TERMINAL

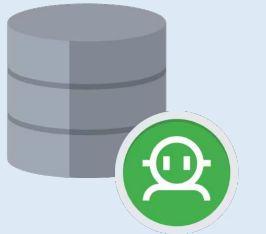
DEMO@dev\_demo >

DEMO | dev | NONE | /Users/rg/DEMO/ | None | No time

**USERNAME | GIT | PROJECT | CWD | TRANSACTION STATUS | TIMING**

> set **statusbar** on

> set **statusbar default** username git project cwd txn timing



## set linenumbers on | off

```
DEMO@dev_demo > set linenumbers on
```

```
DEMO@dev_demo > select 'I really like SQLcl!'  
2* from dual;
```

```
'IREALLYLIKESQLCL!'
```

---

```
I really like SQLcl!
```

```
DEMO@dev_demo > set linenumbers off
```

```
DEMO@dev_demo > select 'But I dont like line numbers'  
from dual;
```

```
'BUTIDONTLIKELINENUMBERS'
```

---

```
But I dont like line numbers
```

```
DEMO@dev_demo > |
```



# set termout on | off

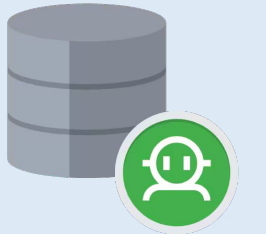
Controls wheter to display the results of SQL when running a script

TERMINAL

```
DEMO@dev_demo > set termout on|
```







## set serveroutput on | off

Print or not output generated by DBMS\_OUTPUT

```
DEMO@dev_demo > set serveroutput on  
DEMO@dev_demo > exec dbms_output.put_line('Hello!');  
Hello!
```

PL/SQL procedure successfully completed.

```
DEMO@dev_demo > set serveroutput off  
DEMO@dev_demo > exec dbms_output.put_line('No message will be shown');
```

PL/SQL procedure successfully completed.

```
DEMO@dev_demo > □
```



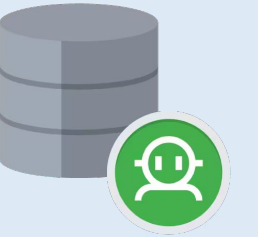
# set feedback on | off

Controls whether to display the number of records returned/affected by a statement

TERMINAL

```
DEMO@dev_demo > set feedback on|
```





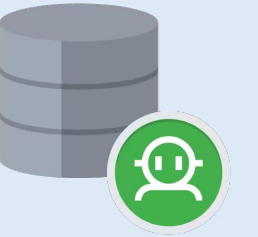
set define on | off | '&' | '#'

Enable / disable prompting

TERMINAL

DEMO@dev\_demo > sh|





# show all params

Display all currently set parameters

```
DEMO@dev_demo > s|
```



Do I need to set those parameters every time?



HOW MANY PIECES DO YOU WISH?



## Define login.sql script

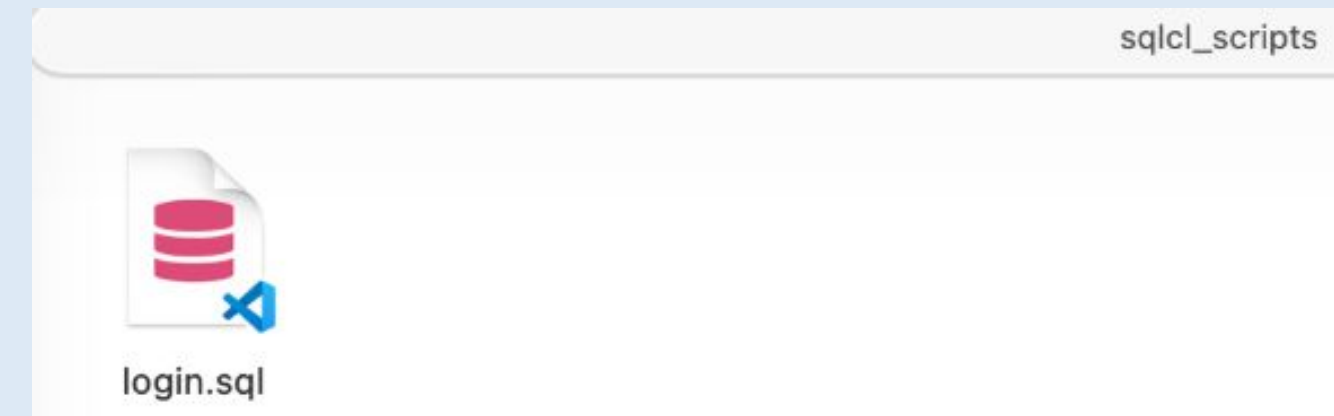


- Add another path into config file -> Users/<your\_user>/ .zshrc (or .bash\_profile)]

```
export SQLPATH="/Users/rg/apps/sqlcl_scripts/"
```

```
.zshrc

export PATH="/Users/rg/apps/liquibase:$PATH"
export PATH="/Users/rg/apps/oracle/sqlcl/bin:$PATH"
export SQLPATH="/Users/rg/apps/sqlcl_scripts/"
export TNS_ADMIN="/Users/rg/apps/oracle/"
```



\* script will be executed at each SQLcl start





## Define custom settings only once

 login.sql 1 X

Users > rg > Downloads >  login.sql > ...

```
1  set sqlprompt "@|blue _USER|@@|green _CONNECT_IDENTIFIER|@ > "  
2  set linenumbers off  
3  set define '&  
4  set feedback on  
5  set serveroutput on  
6  set termout on  
7  set sqlformat ansiconsole  
8  set statusbar on  
9  set statusbar default username git project cwd txn timing
```

# Q&A

---



## Audience Q&A







**GitHub - rgrzegorzcyk/conferences: A place  
for public content related to my  
presentations at various conferences**

github.com

<https://github.com/rgrzegorzcyk/conferences>

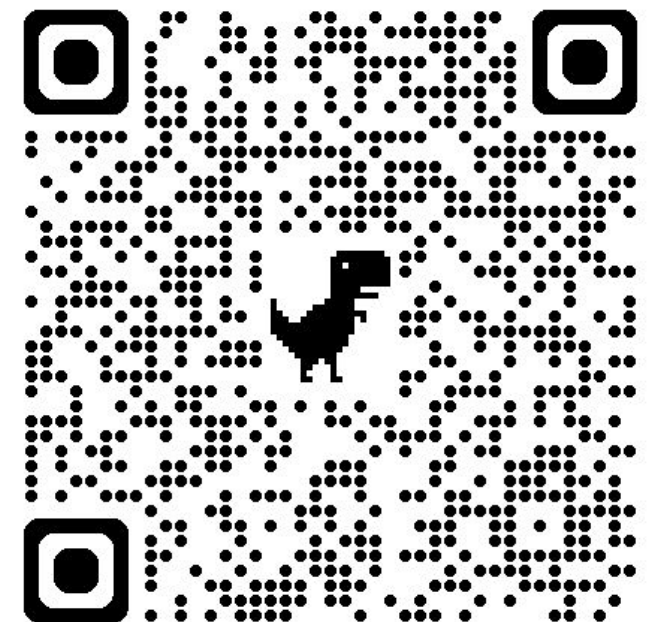


# Oracle SQLcl: What Every APEX & PL/SQL Dev Should Know (DBAs, You're Invited too!)

Updated June 8, 2026 • 3 min read



My blog post article – hub for all things SQLcl» [here](#)



# Thank You For Attending!

Please complete the session survey  
in the conference app.



Rafal Grzegorzczak  
rgrzegorzczak@pretius.com

rgrzegorzczak  <https://rafal.hashnode.dev/>



PROBLEMS OUTPUT TERMINAL CHAT

DEMO@dev\_demo > |

